web3

Two thousand
twenty two

gradients so **based**

State *of* Crypto

An overview report

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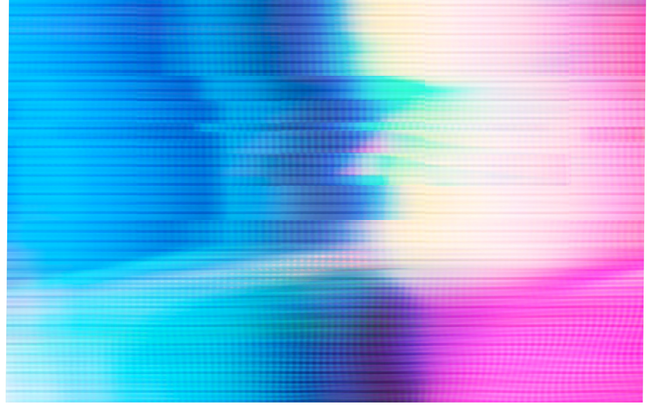

What **is web3?**

01.

The first era of the modern internet — roughly 1990-2005 — was about open protocols that were decentralized and community-governed. Most of the value accrued to the edges of the network: users and builders.

02.

The second era of the internet — roughly 2005-2020 — was about siloed, centralized services. Most of the value accrued to a handful of large tech companies.

03.

We are now beginning the third era of the internet — what many call web3 — which combines the decentralized, community-governed ethos of the first era with the advanced, modern functionality of the second era. This will unlock a new wave of creativity and entrepreneurship.

Source: https://cdixon.mirror.xyz/0GBv9WRI-LQE6F4r2gGczG6AVeFINuvt3HETHZ5A93U

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Why Web3 *Matters*

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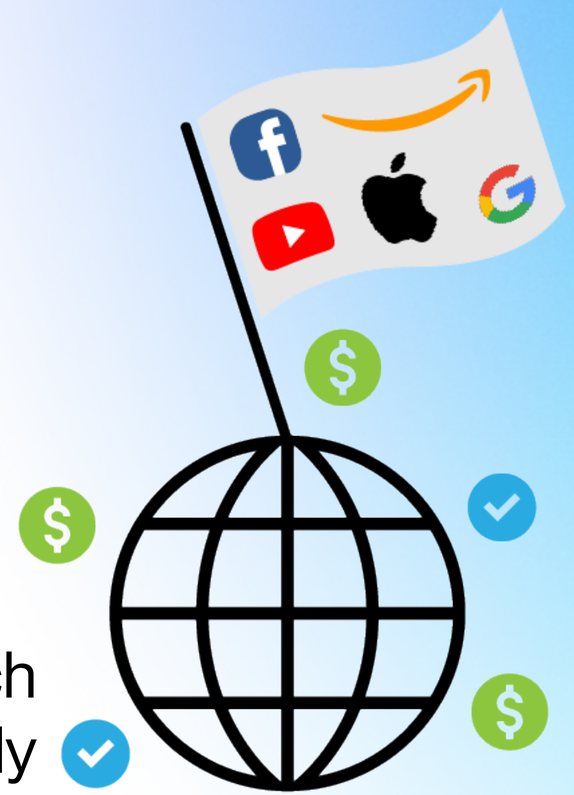
Digital
Authoritarianism



Big Tech
Oligopoly

The internet as we know it *is flawed*

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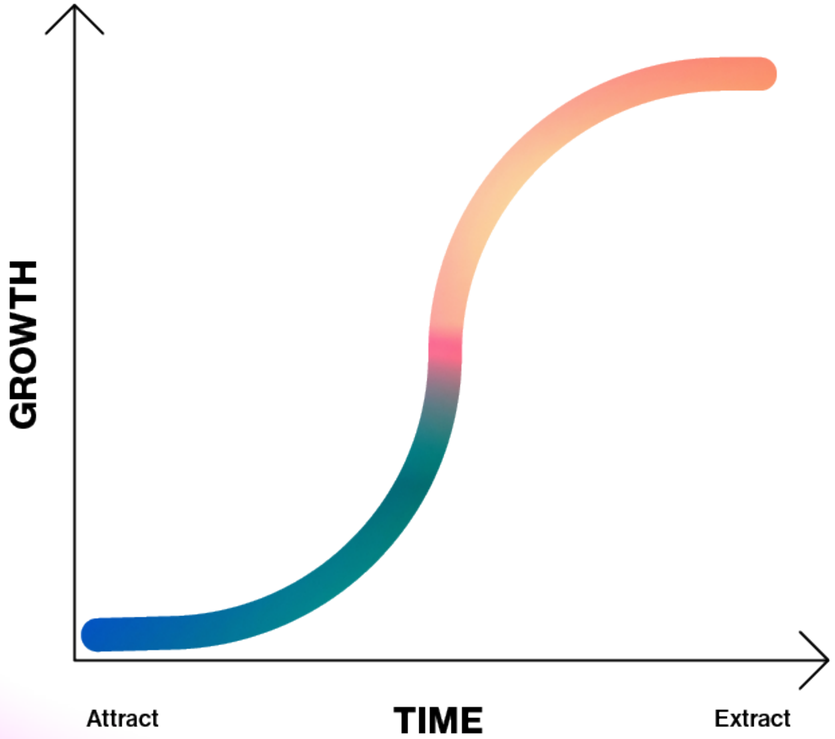
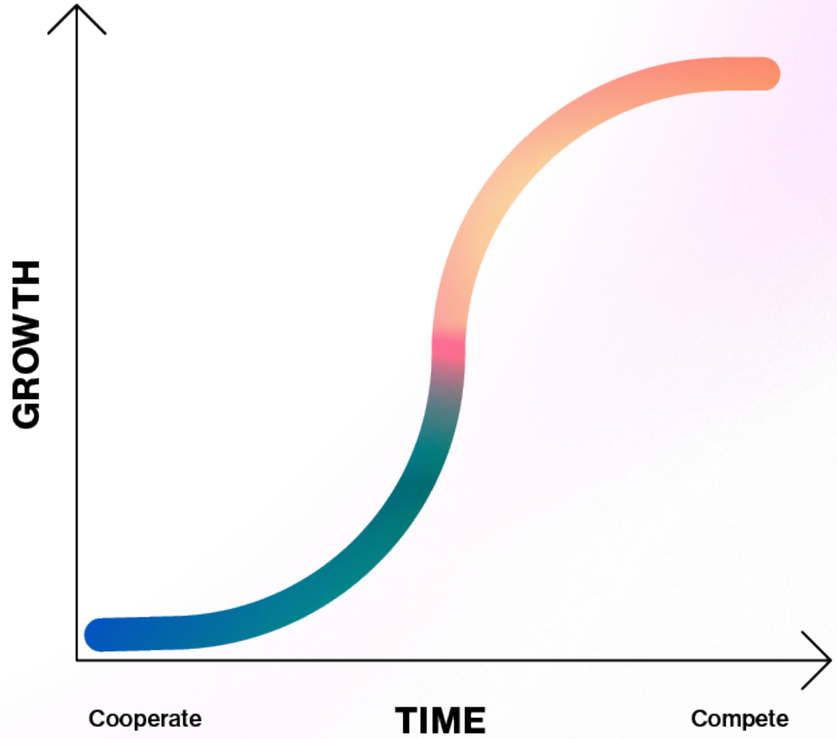
After friendly beginnings, web2 companies have become *more extractive and less cooperative*





Charts for illustrative purposes only

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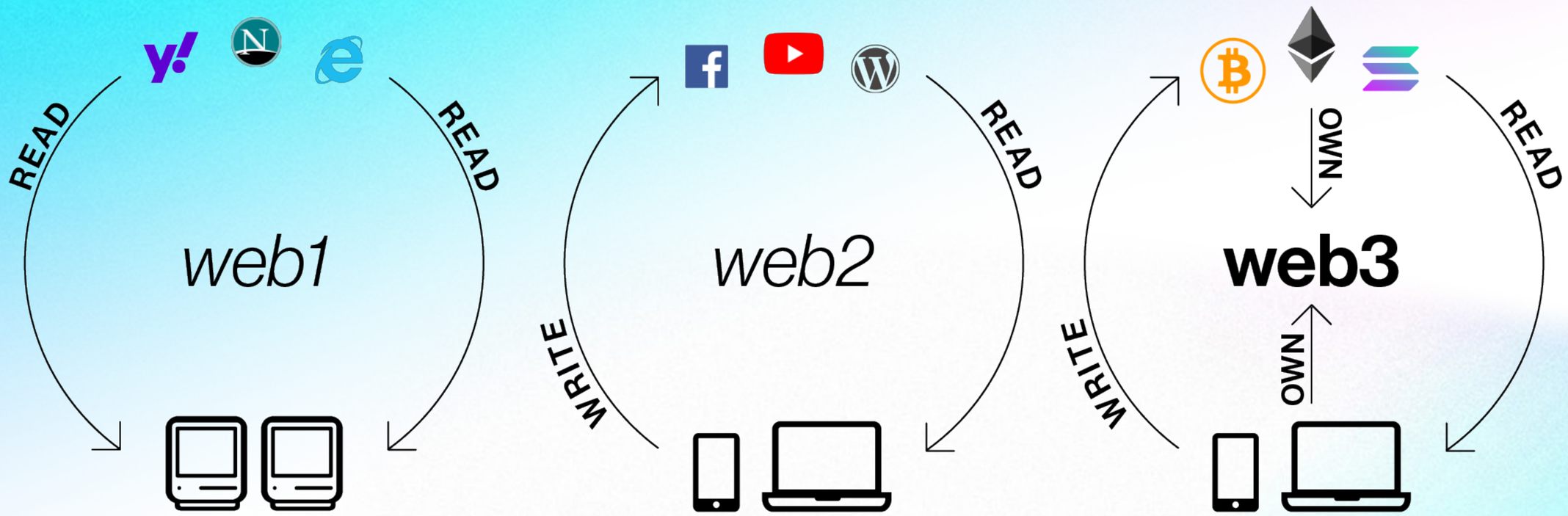

Web3 gives people property rights: the *ability to own* a piece of the internet

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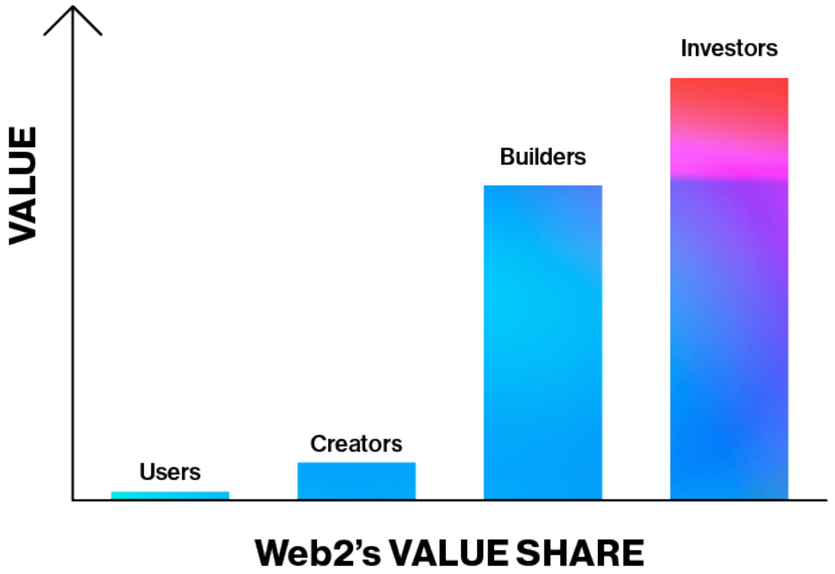
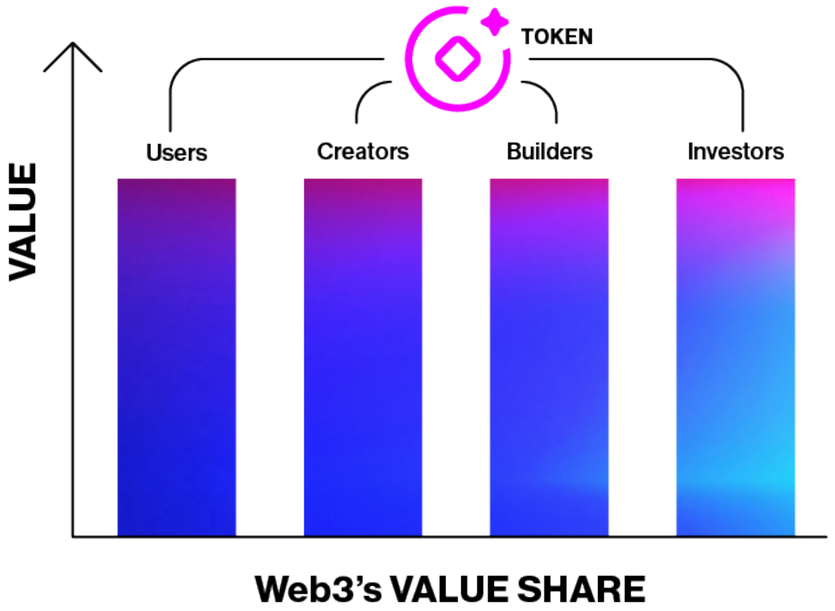
Web3 aligns network participants to work together toward a common goal — *the growth of the network*





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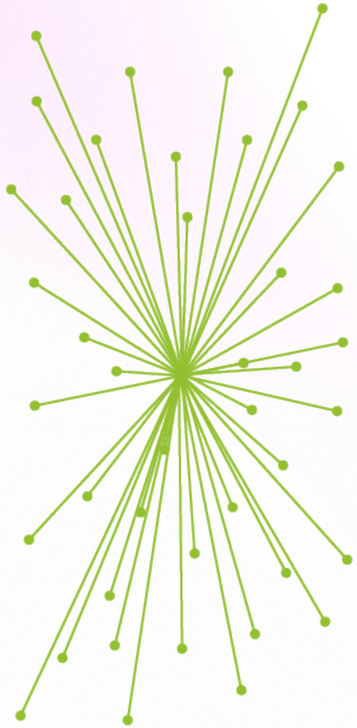
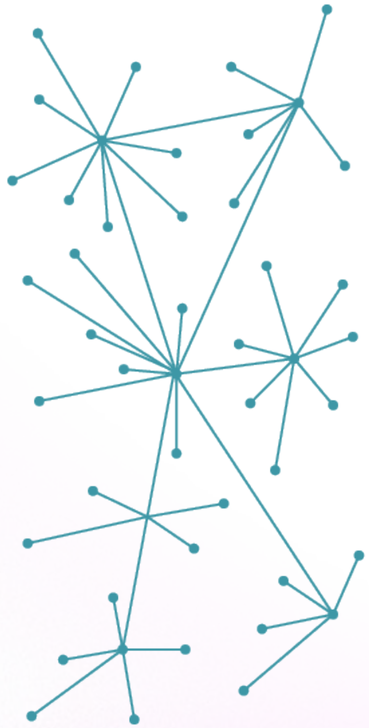
Web3 empowers a collective owned future over a corporate or government owned future

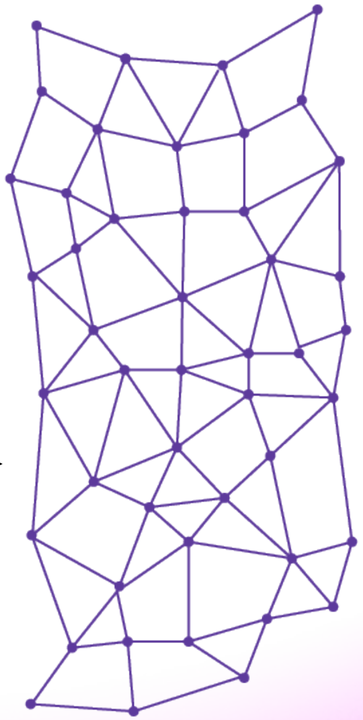
Source: RAND





◀ **Single points of failure & control**

Resilient, equitable & participatory ▶



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Cycles *of* Innovation

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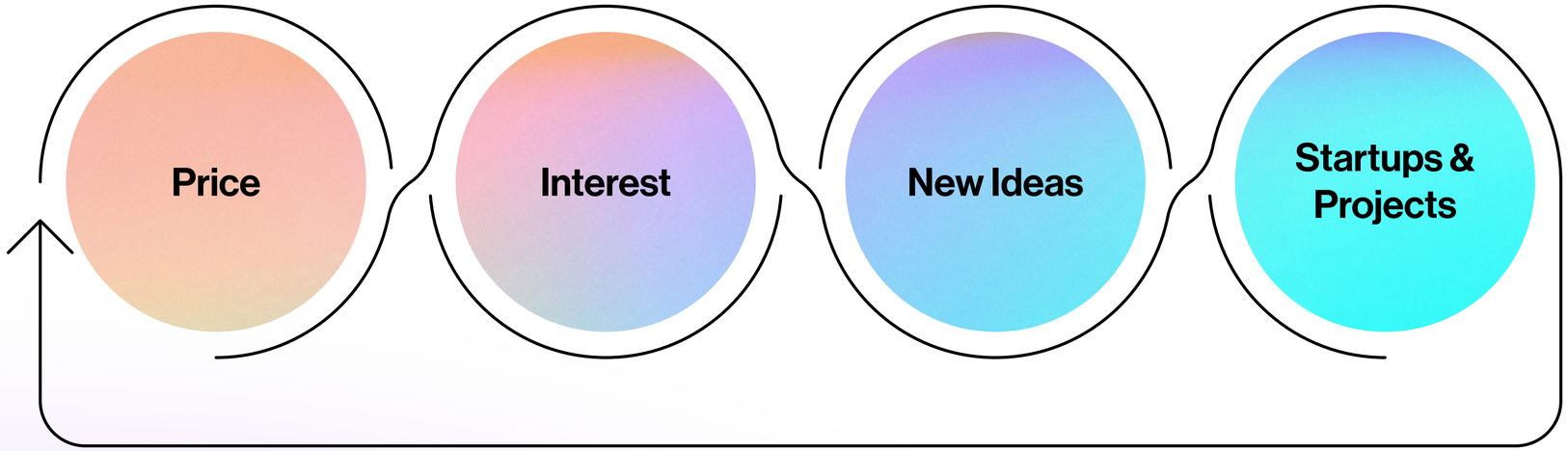
The crypto market
develops in cycles



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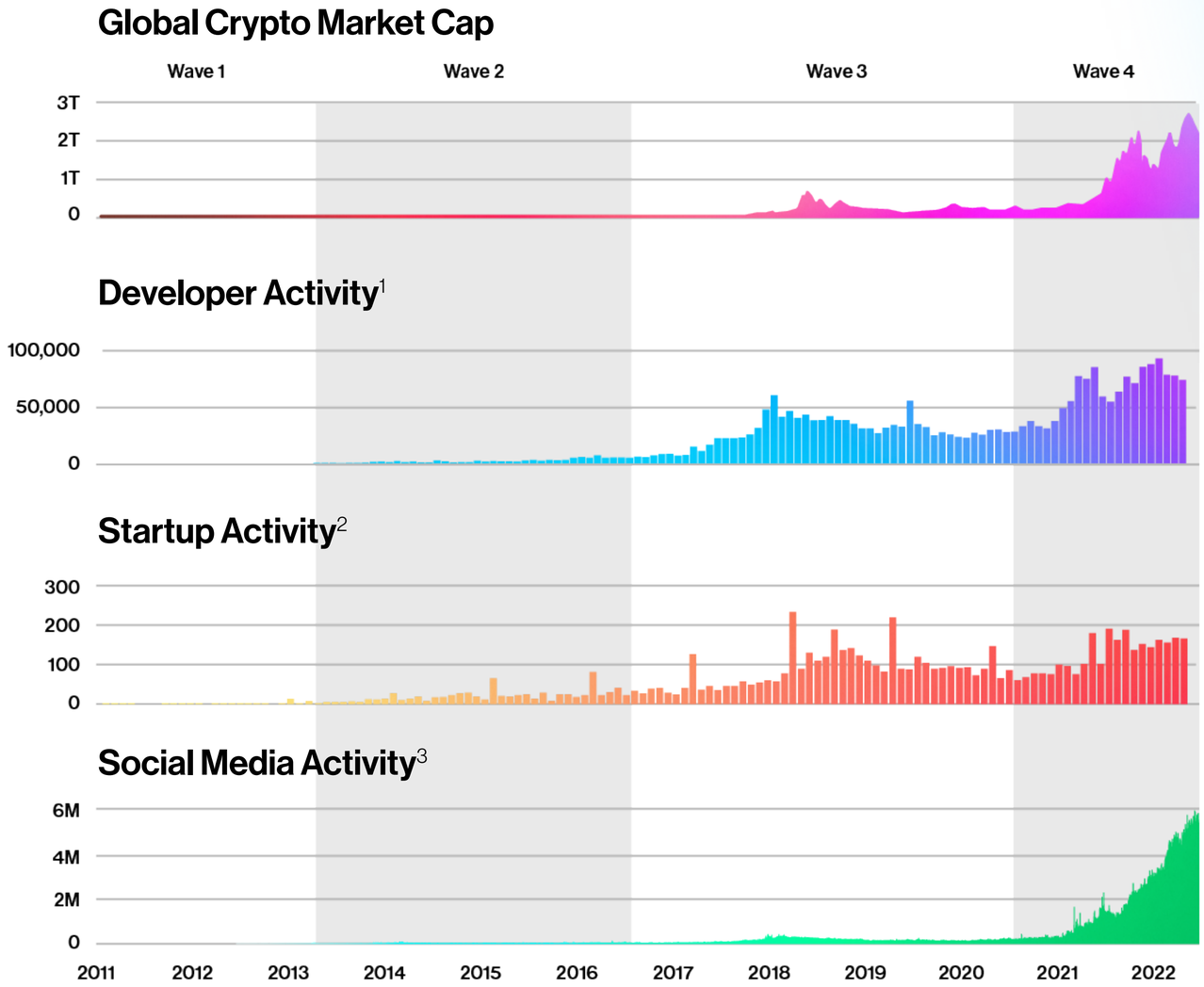
These cycles appear chaotic from the outside **but** *have an underlying order*



Source: CoinMarketCap, GitHub, Pitchbook, Twitter

Data is as of 5/12/2022

1 Developer activity based on stars of crypto-related repositories on Github.

2 Startup activity based on crypto-related initial funding rounds via Pitchbook

3 Social media activity based on crypto-related comments on Twitter

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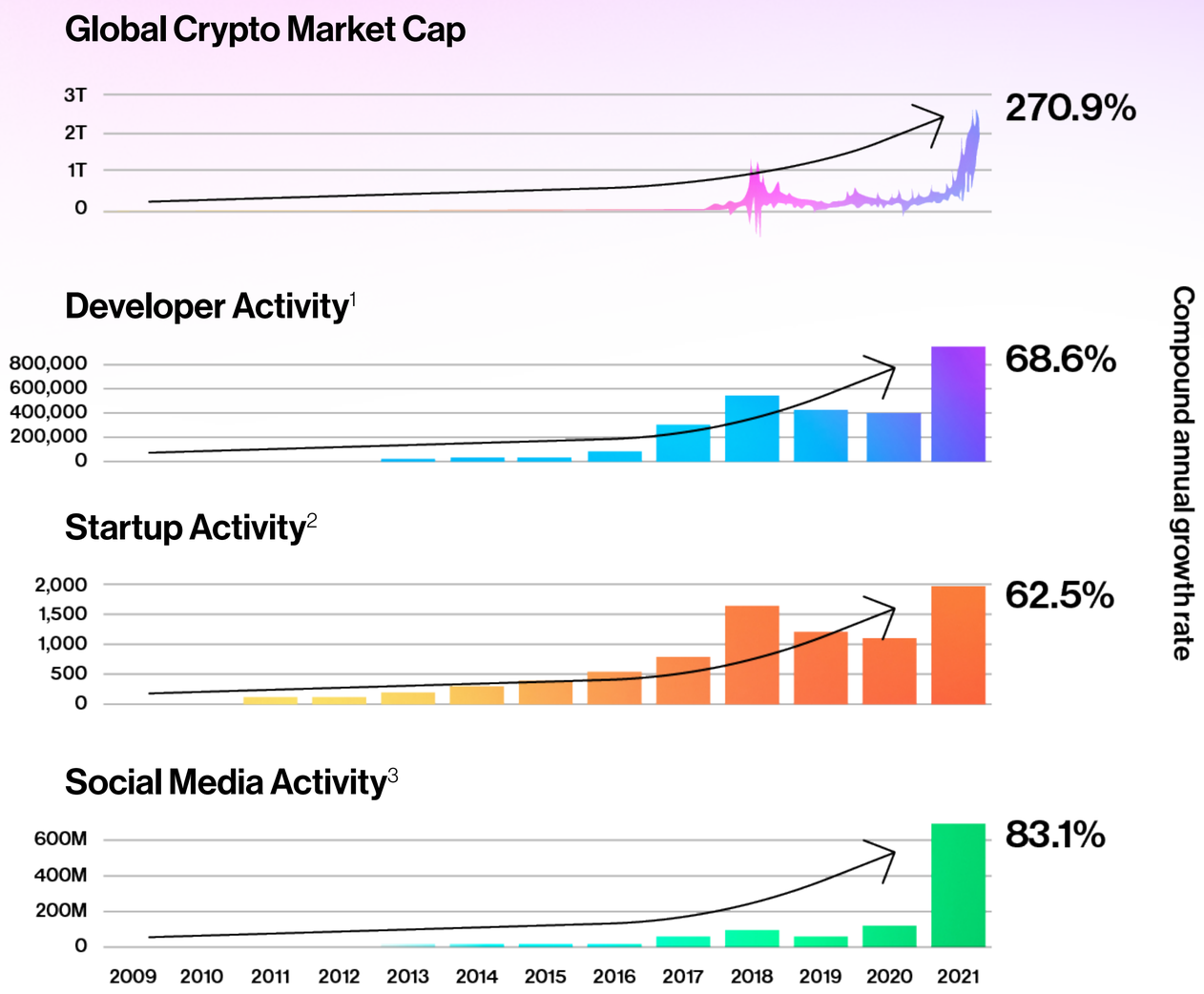
The result is *consistent long-term growth,* driven by a feedback loop between interest and innovation



Source: CoinMarketCap, GitHub, Pitchbook, Twitter

Data is as of 12/31/2021

1 Developer activity based on stars of crypto-related repositories on Github.

2 Startup activity based on crypto-related initial funding rounds via Pitchbook

3 Social media activity based on crypto-related comments on Twitter

Layer *1*
Blockchains

2022

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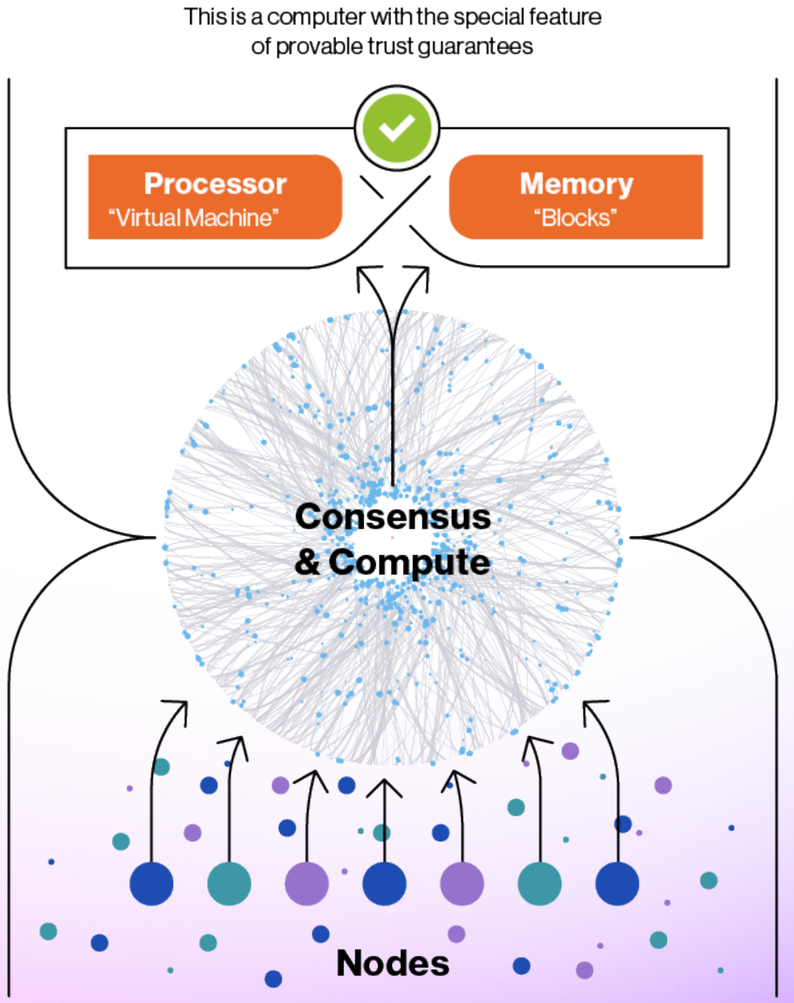
Blockchain computers are composed of nodes: physical computers, like PCs, joined together *via a consensus mechanism*



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Web3 is multi-chain; **developers have many options when it comes to building on blockchains**

	Ethereum	Solana	Polygon	BNB Chain	Avalanche	Fantom
Active Addresses	5.5M	15.4M	2.6M	9.4M	609K	308K
Daily Transactions	1.1M	15.3M	3.4M	5.0M	832K	743K

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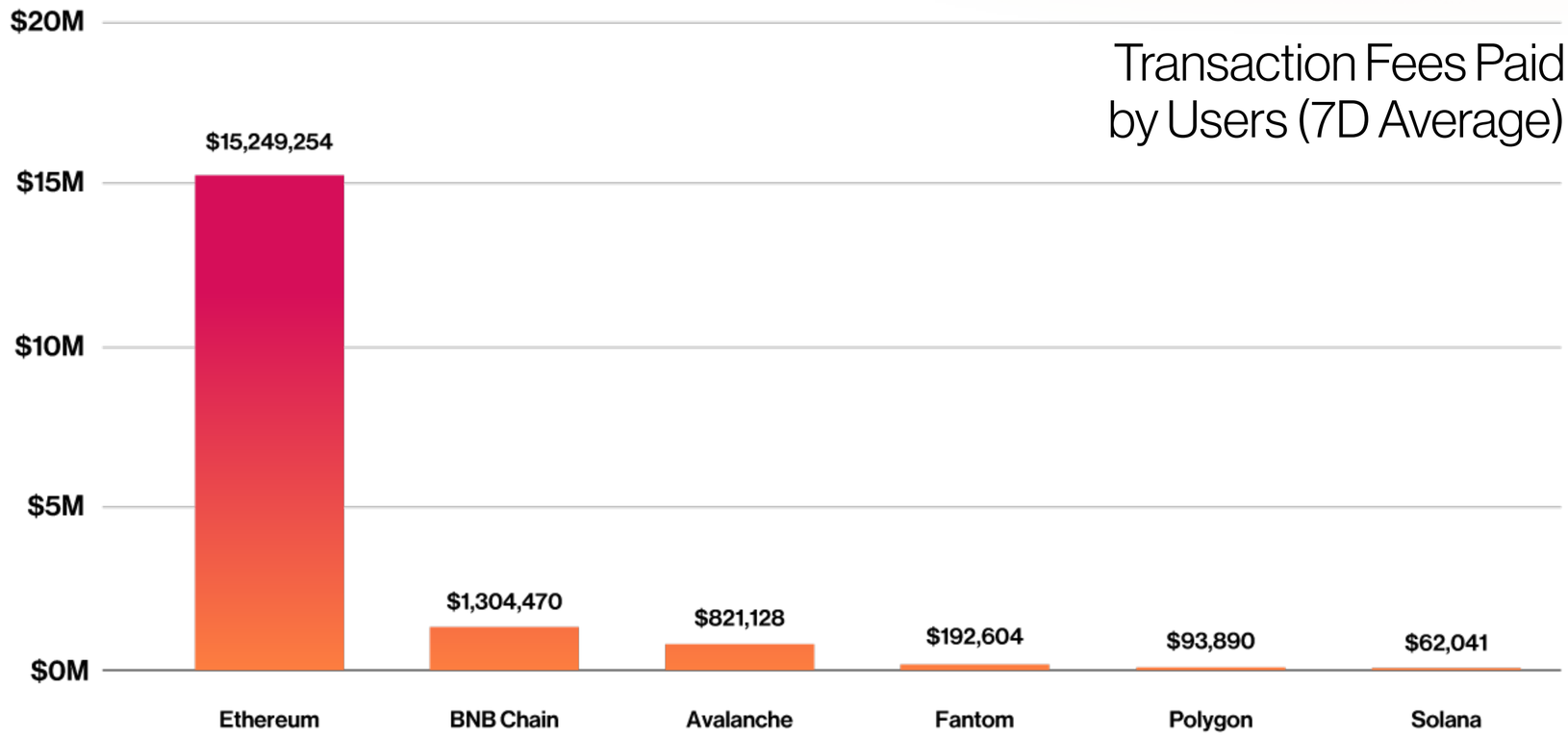
Source: Nansen; Active addresses is measured over a 30-day period as of 5/12/2022; Daily transactions is represented by a 30-day average as of 5/12/2022.

While other smart contract platforms rival Ethereum in users and usage, *the demand for block space is unmatched*



Source: https://cryptofees.info; Data is as of 5/12/2022.

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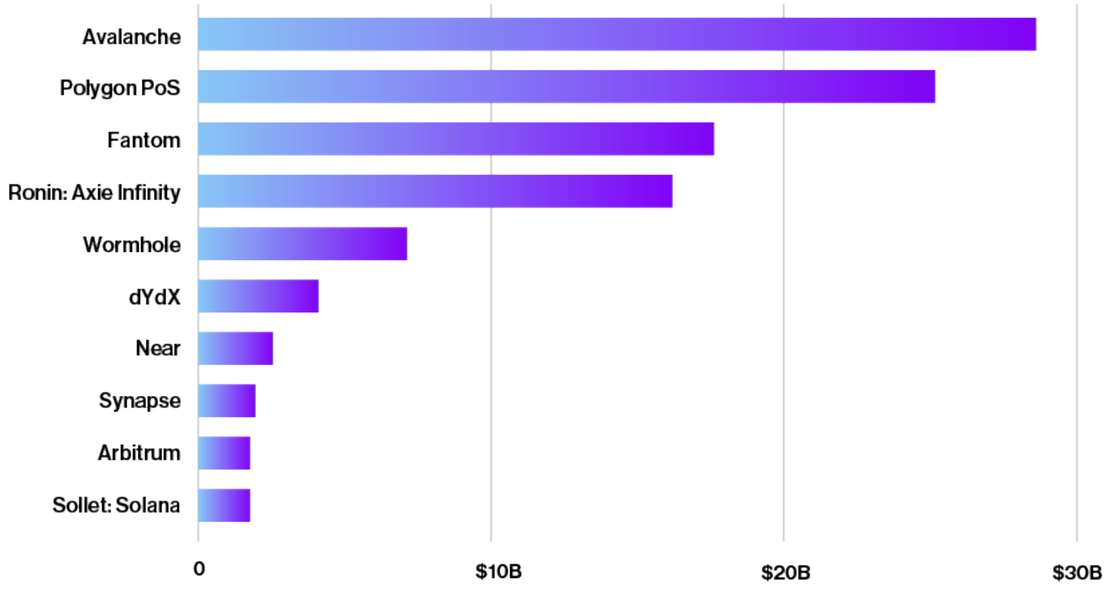
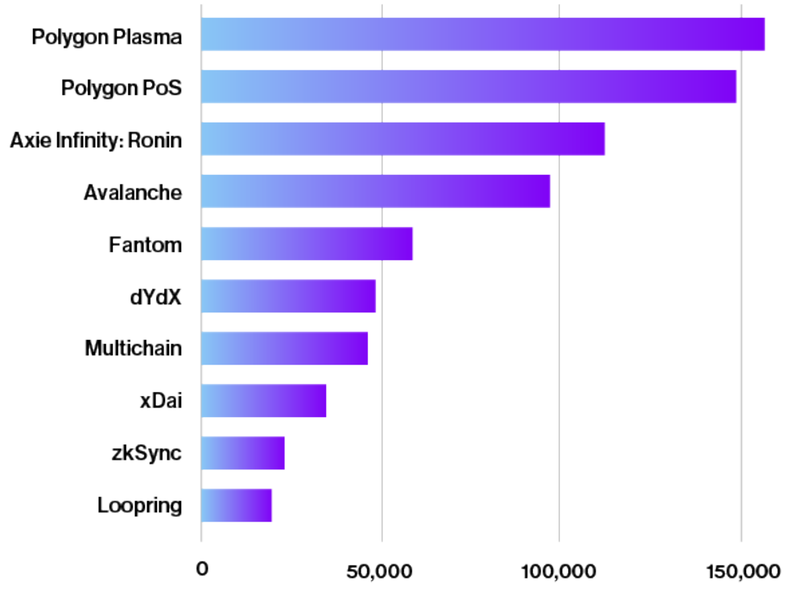
With Ethereum L1 as the hub, a significant amount of value is being *deposited into bridges*

Top Ethereum Bridges by Total Volume In



Top Ethereum Bridges by Total Depositors



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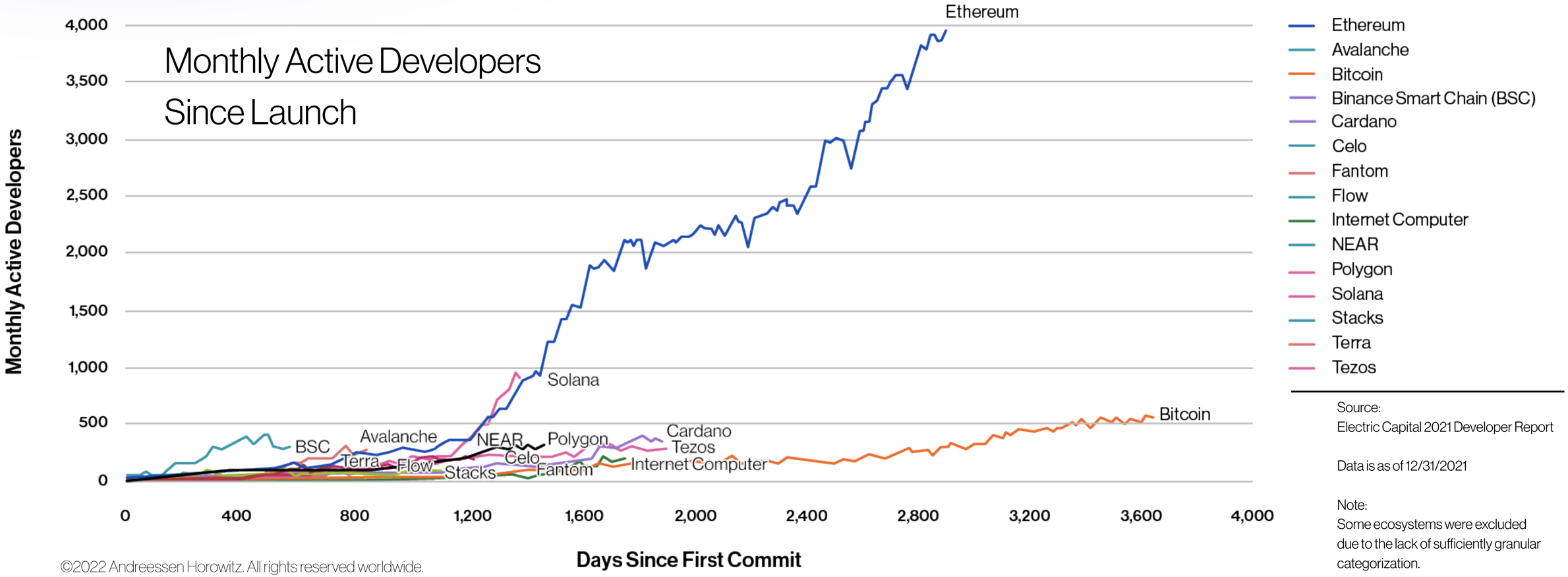
Source: Nansen; Includes bridges with > 10k depositors; Data is as of 5/12/2022.

Ethereum continues to attract the most developers, **but several emerging ecosystems are on a similar trajectory**



Source:
Electric Capital 2021 Developer Report

Data is as of 12/31/2021

Note:
Some ecosystems were excluded due to the lack of sufficiently granular categorization.

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Layer *2* Scaling

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A layer 2 is a separate blockchain that extends the base layer **and** *inherits its security guarantees*

Optimistic Rollups

Transactions are assumed to be valid, but can be challenged if necessary

- More production-ready
- Correctness relies on game-theoretical incentives
- Finality on L1 depends on a 1-week challenge period
- Easier to program
- EIP-4844 (proto-danksharding) can make these rollups very cheap for users

Optimism

ARBITRUM

Zero-Knowledge Rollups

State transitions are computationally verified off-chain using validity proofs

- Less production-ready
- Correctness is mathematically guaranteed
- Finality on L1 is instant
- Harder to program
- Proofs can be computationally expensive, though this is rapidly improving

zkSync

STARKNET

Aztec

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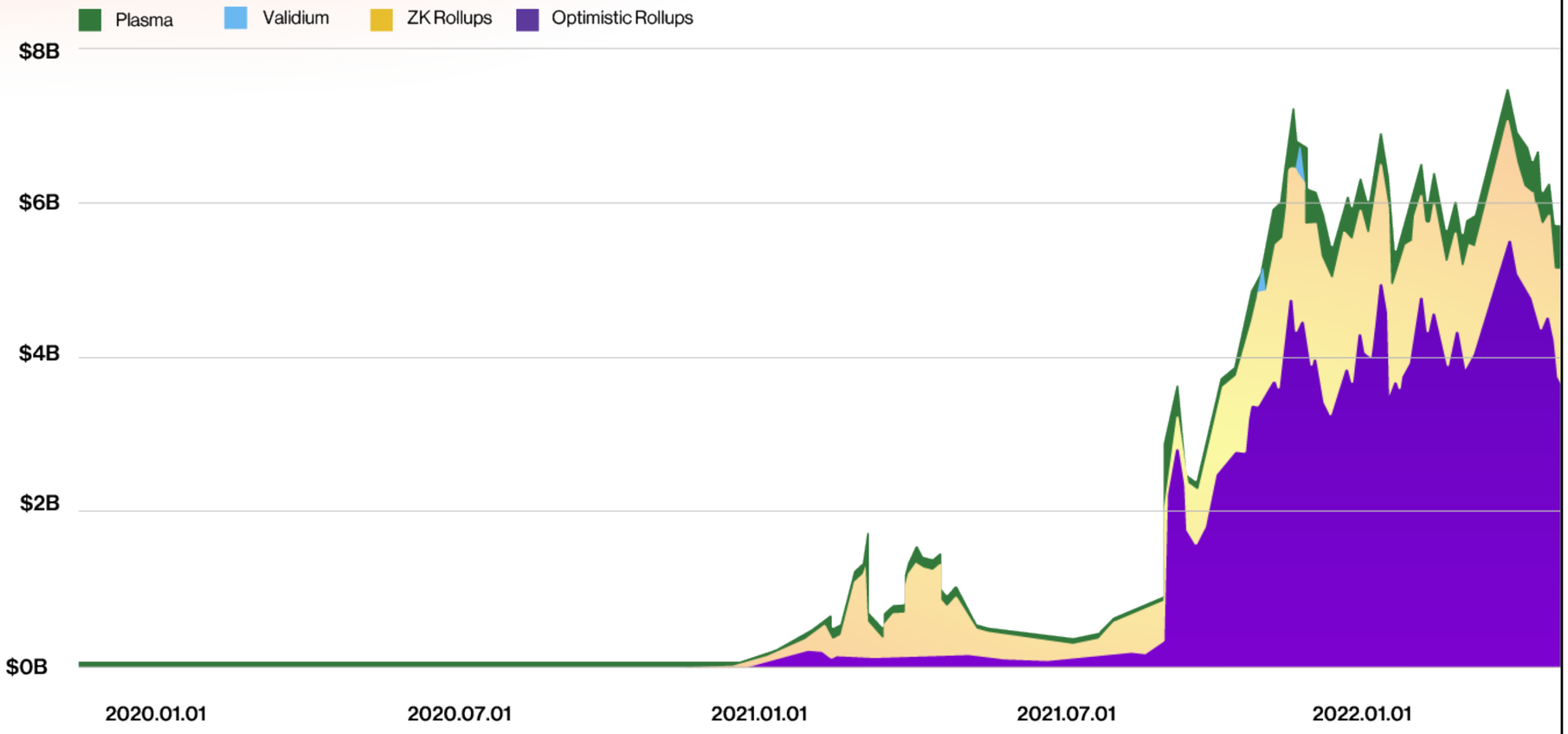
Optimistic rollups are the most popular L2 tech today, but *ZK rollups are just beginning to come online*

Total Value Locked on Ethereum Layer 2s



Source: https://l2beat.com/ Data is as of 5/12/2022.

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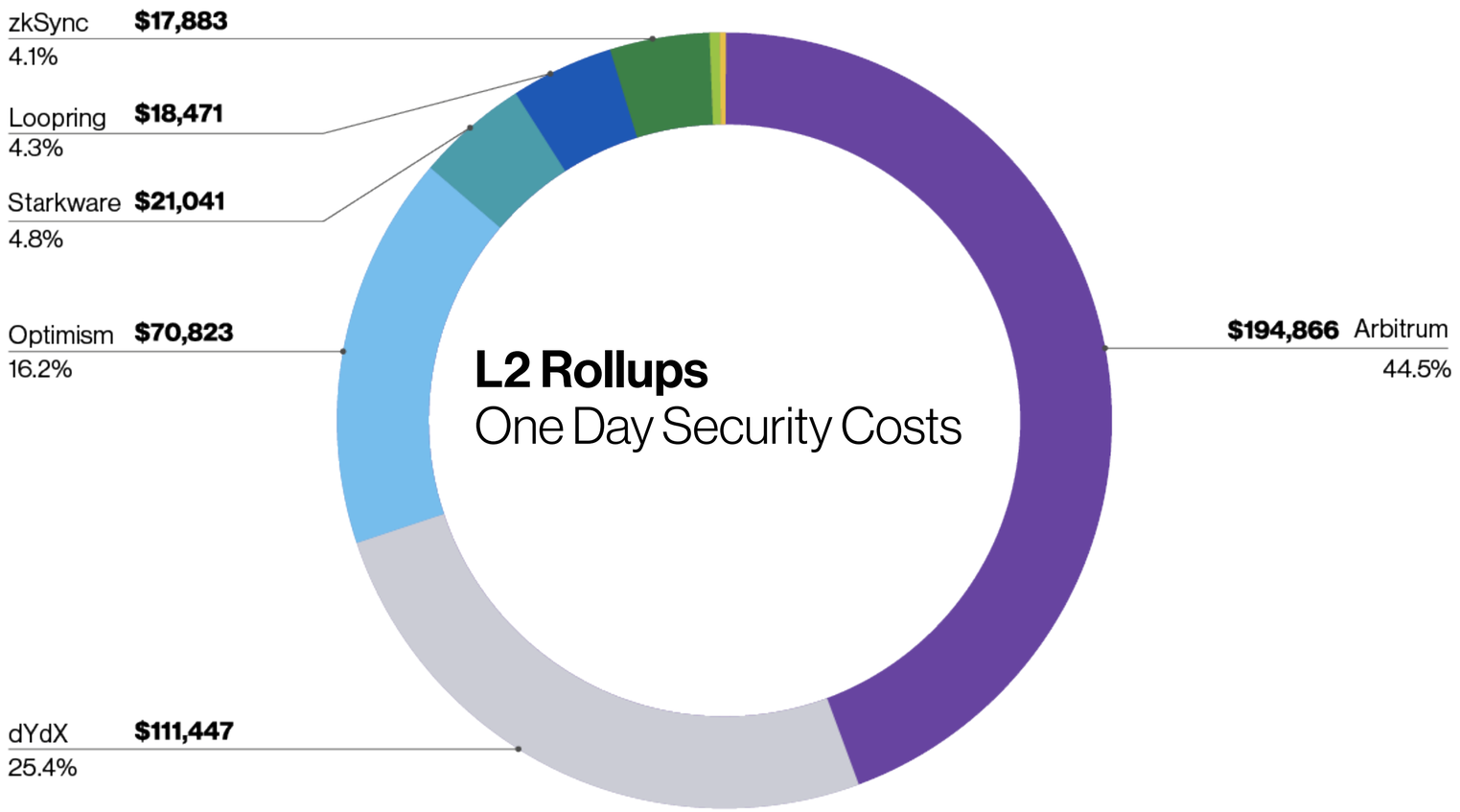
L2 rollups pay for *Ethereum's security*

Today, L2 rollups contribute to

~1.5%

of all fees paid on Ethereum



Source: https://l2fees.info/; Data is as of 5/12/2022.

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L2 rollups are competing hard against each other to *bring transaction fees down*

Name	Send ETH	Swap Tokens
Metis Network	$0.01	$0.07
Loopring	$0.20	$0.48
zkSync	$0.22	$0.54
Polygon Hermez	$0.25	-
Boba Network	$0.72	$0.94
Optimism	$0.88	$1.30
Arbitrum One	$2.27	$3.15
Aztec Network	$3.09	-
Ethereum	**$4.48**	**$22.41**

Source: https://l2fees.info/
Data is as of 5/12/2022.

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DeFi

2022

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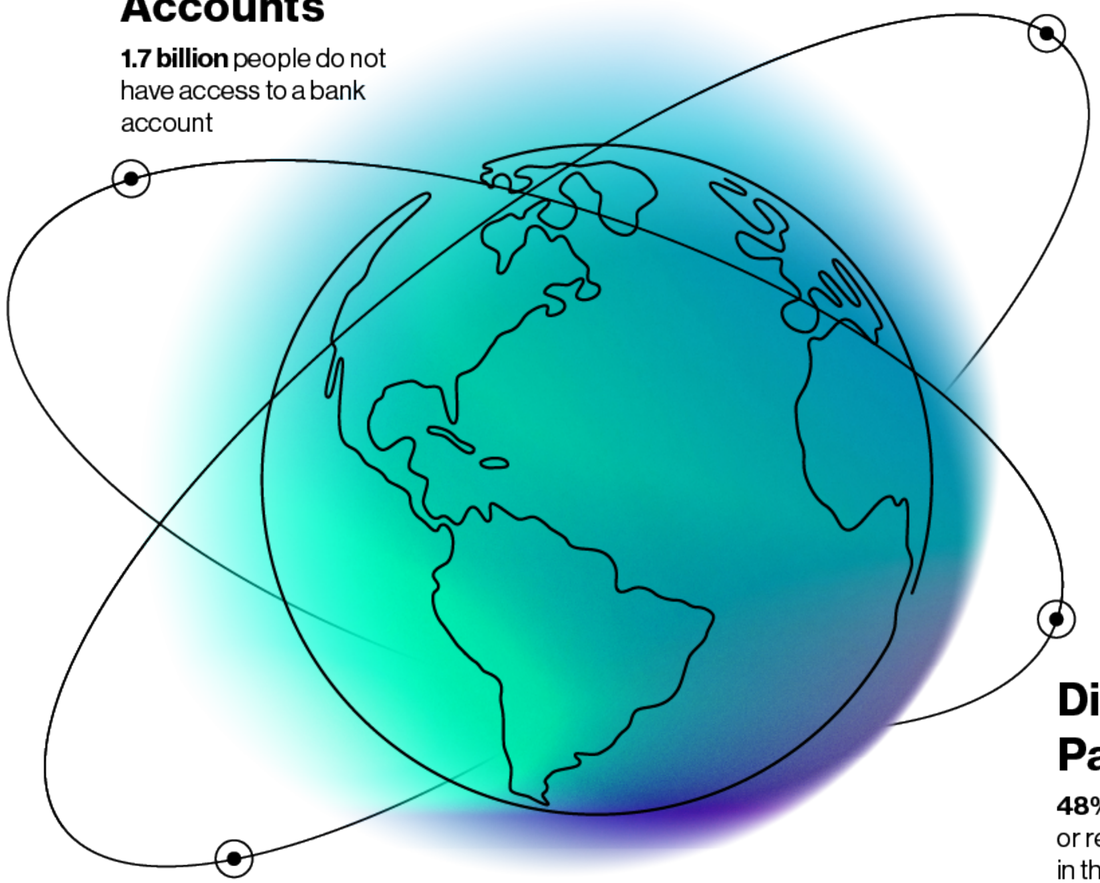
A huge part of the world **is** *underserved by existing financial institutions*



Bank Accounts

1.7 billion people do not have access to a bank account

Remittance Payments

$650B in global remittances annually, with an average cost of **6%**

Digital Payments

48% of adults have not sent or received digital payments in the past year

Mobile + Internet

Of the unbanked, **1 billion** have access to a mobile phone and **480 million** have internet

Sources:
The Global Findex Database, 2017 Report;
The World Bank (2020 and 2021 data)

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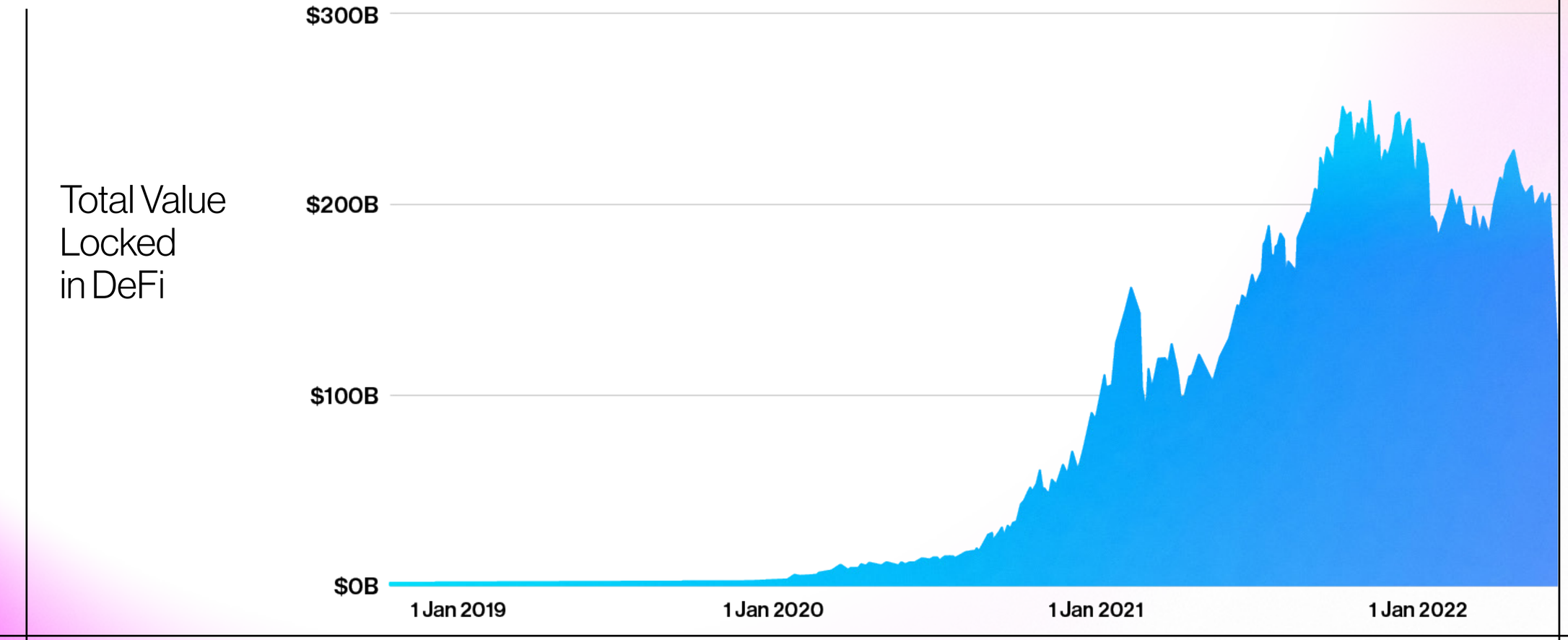
DeFi has grown from nearly zero to over $100 billion in *less than two years*



Source: Defi Llama; Data is as of 5/12/2022.

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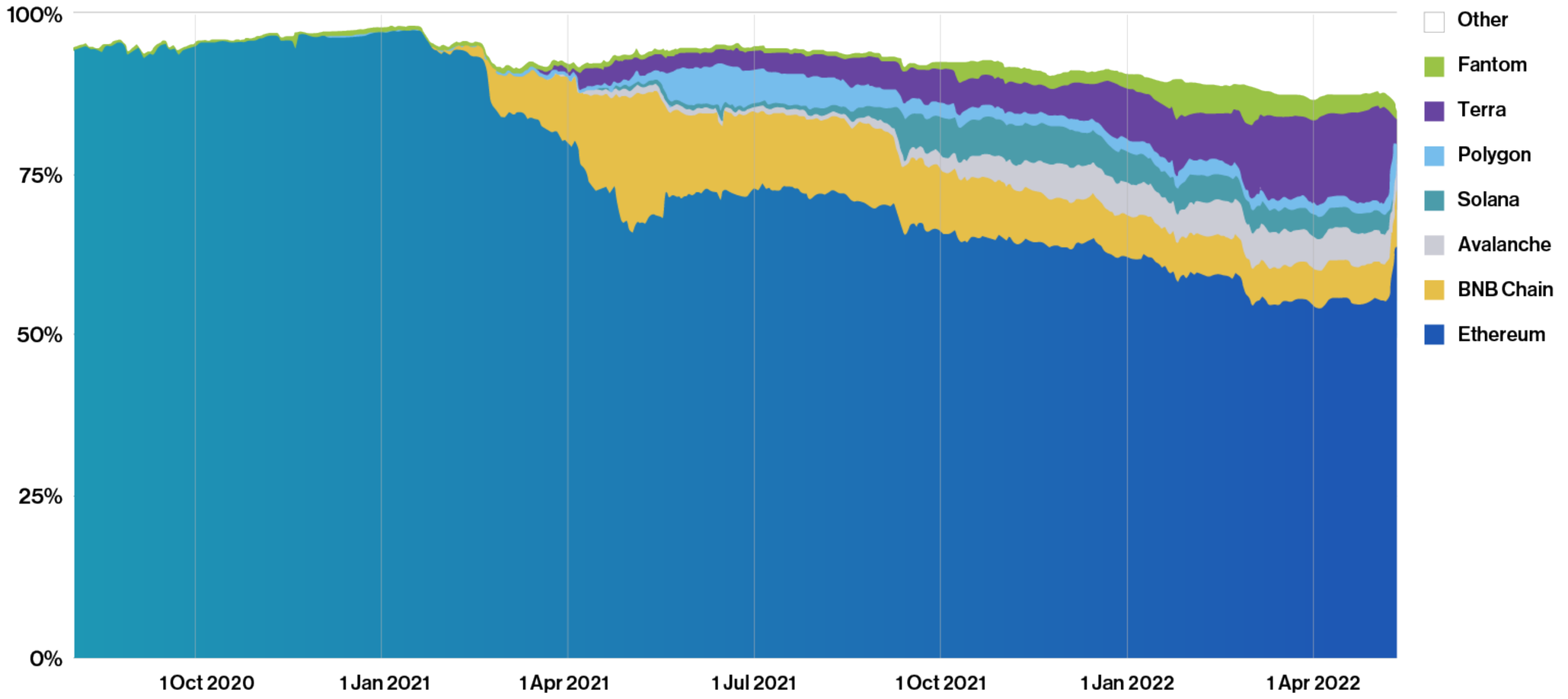
Newer blockchains are trying to replicate the *success of DeFi* on Ethereum



DeFi Market Share (TVL) by Blockchain

Source: Defi Llama;
Data is as of 5/12/2022.

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Token exchange and lending protocols are the *most popular DeFi use cases* today

Source: Token Terminal
Data is through April 2022.

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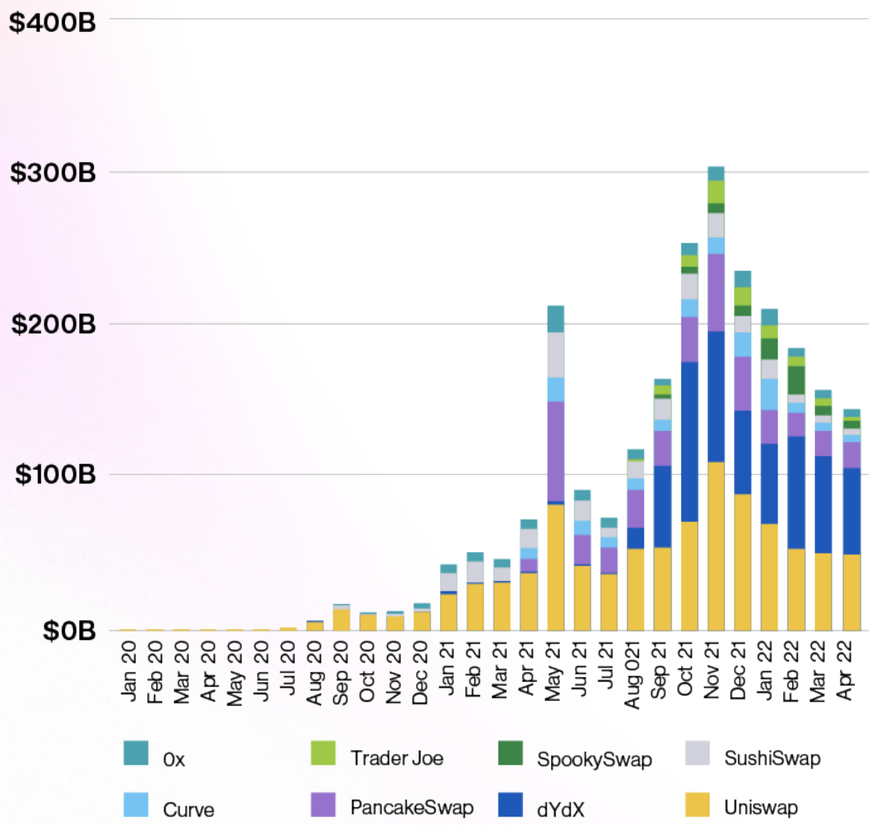
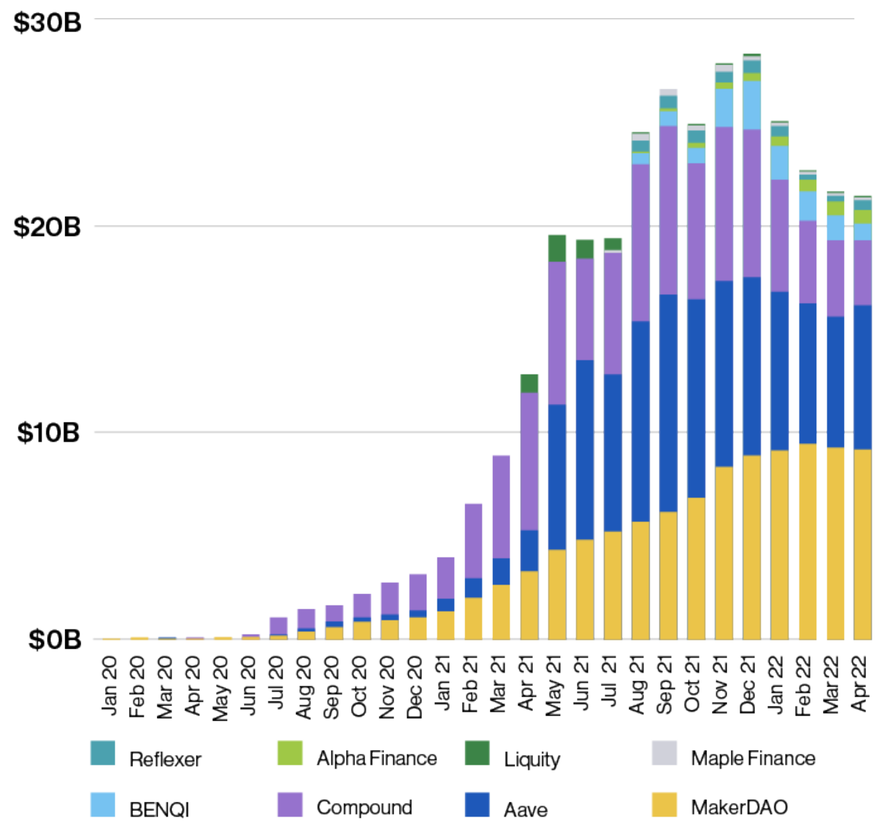
Trading Volume on Decentralized Exchanges



Borrowing Volume on Decentralized Lending Protocols



Proof-of-stake **is coming to Ethereum**

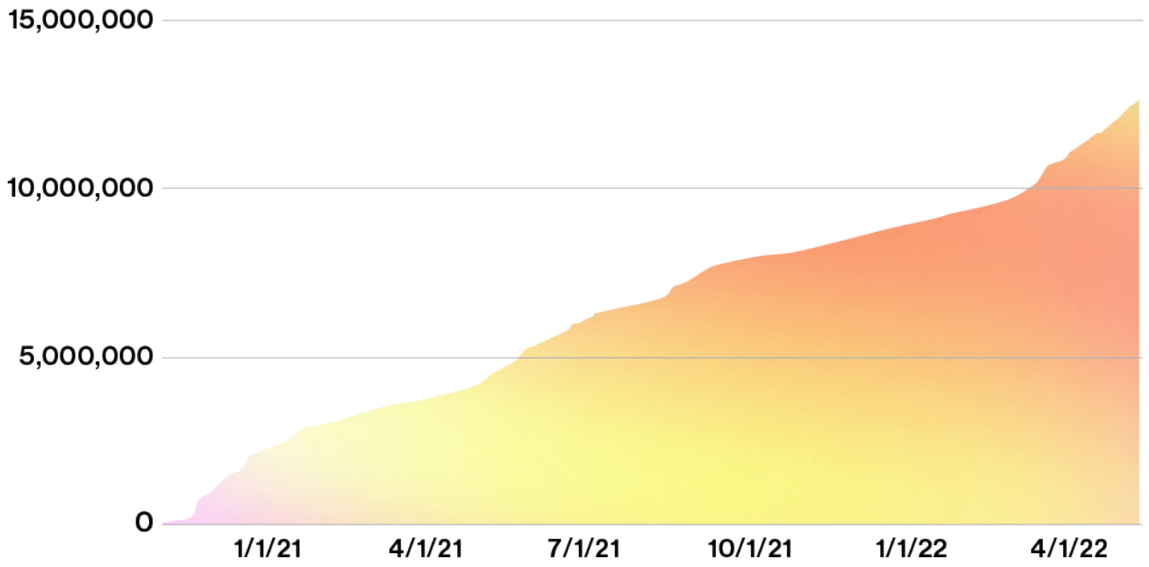
Source: Nansen
Data is as of 5/12/2022.

12.6M ETH
Total ETH Deposited

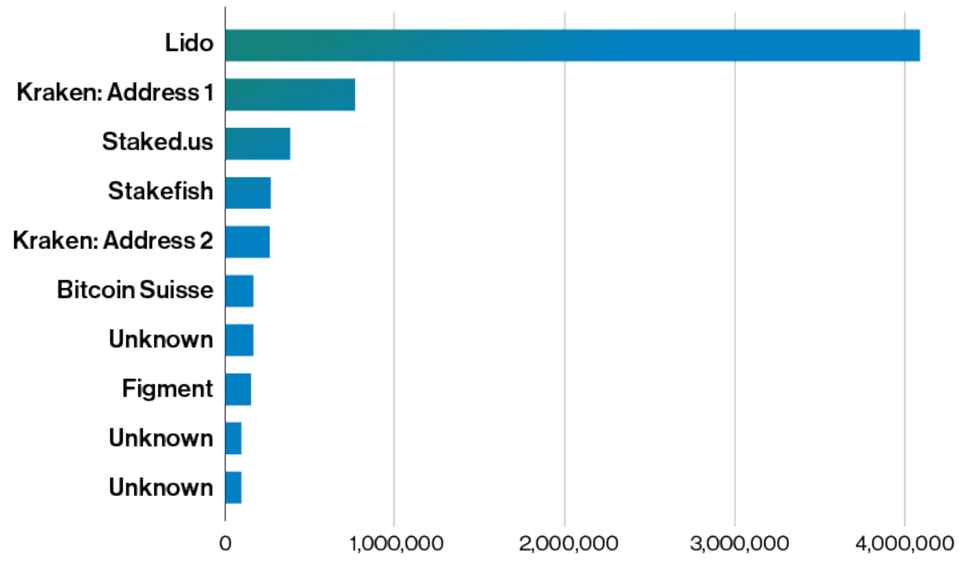
73K
Unique Depositors

392K
Unique Validators

Total ETH Staked to Beacon Chain



Top ETH 2 Deposit Addresses



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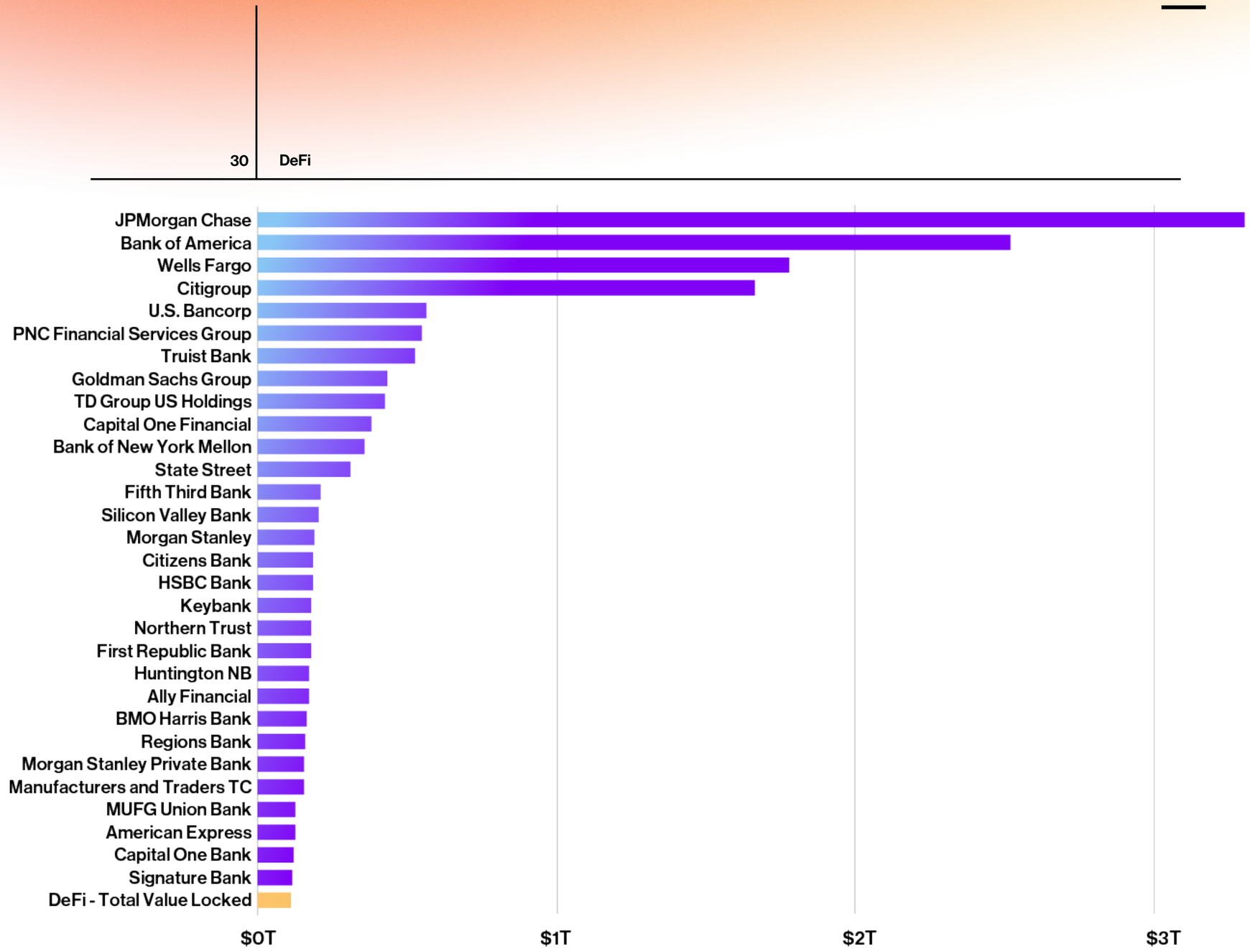
DeFi would represent the *31st largest US bank* by total assets under management



Source:
Defi Llama, https://www.federalreserve.gov/releases/lbr/current/;
Bank AUM data is as of 12/31/2022
DeFi TVL data is as of 5/12/2022.

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Stablecoins

2022

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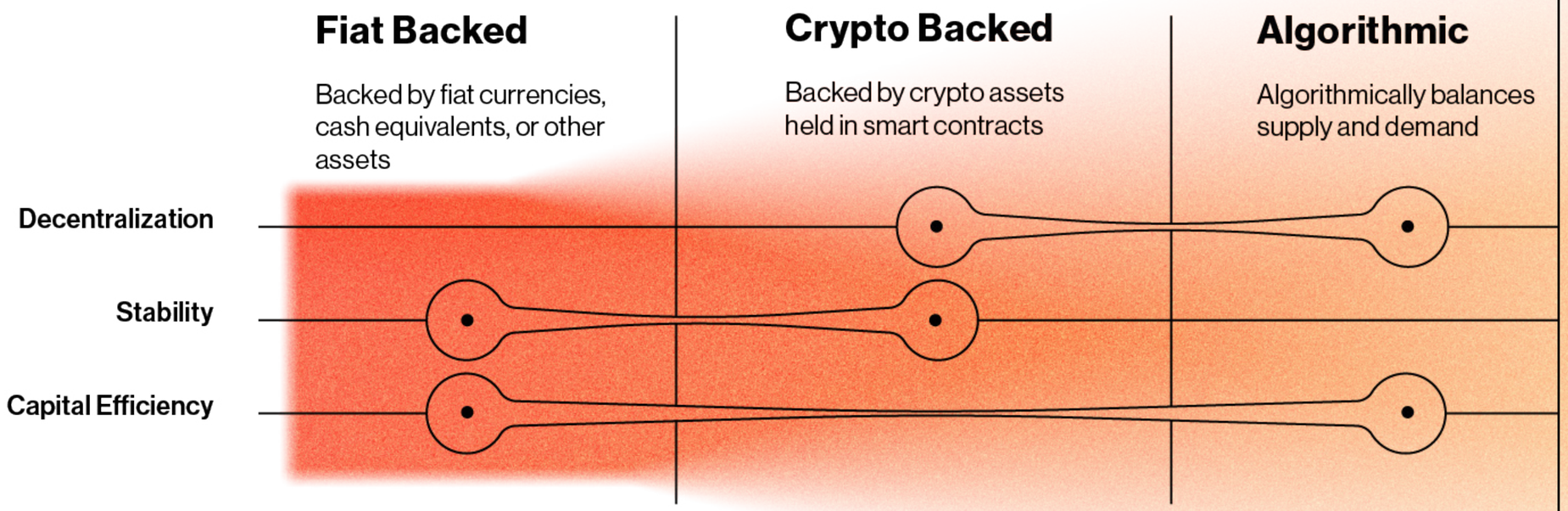
There are *three primary categories* of stablecoins, though the lines between them are blurry



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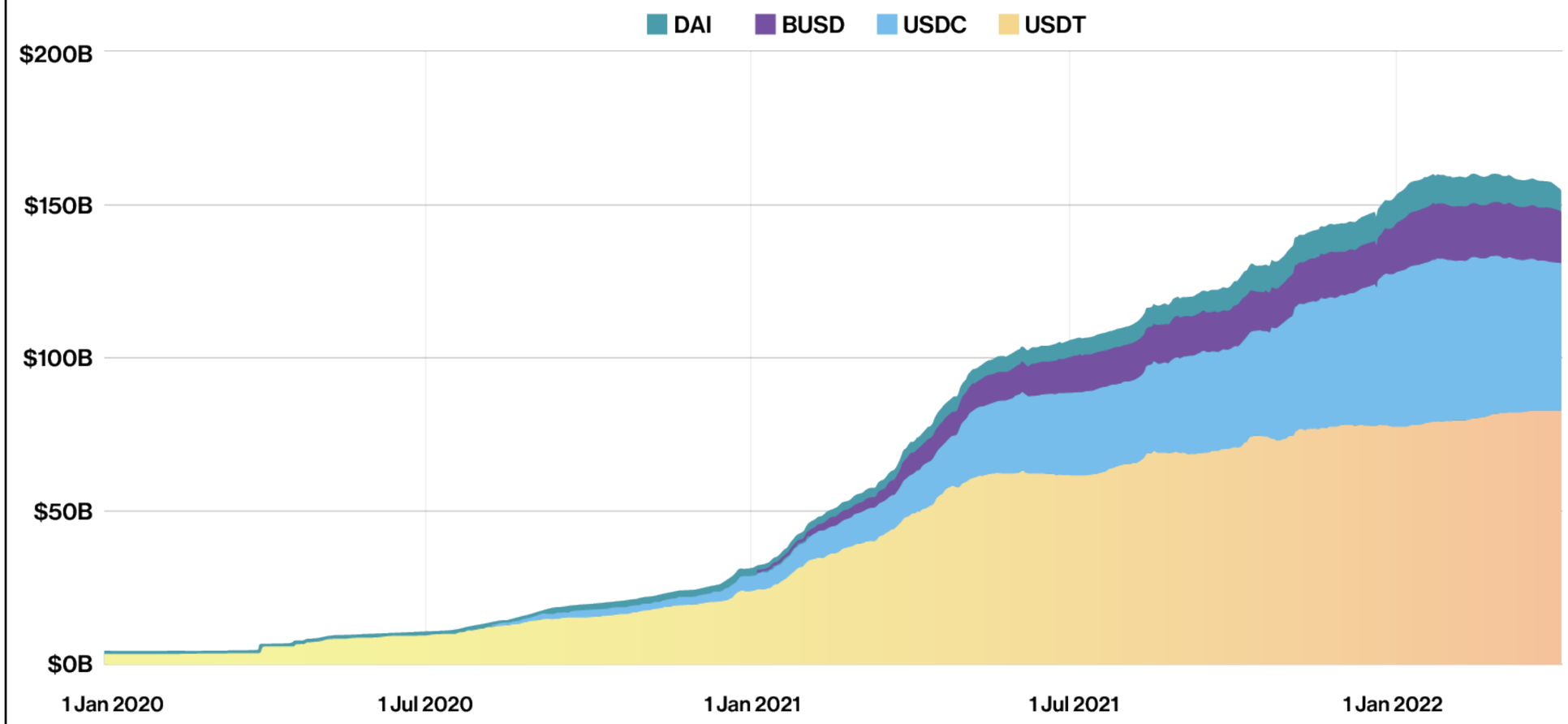
Increasing use of stablecoins demonstrates demand for a digital dollar

Stablecoins

Total Stablecoin Supply



Source: CoinGecko
Data is as of 5/12/2022.

Only includes stablecoins that are believed to be fully collateralized.

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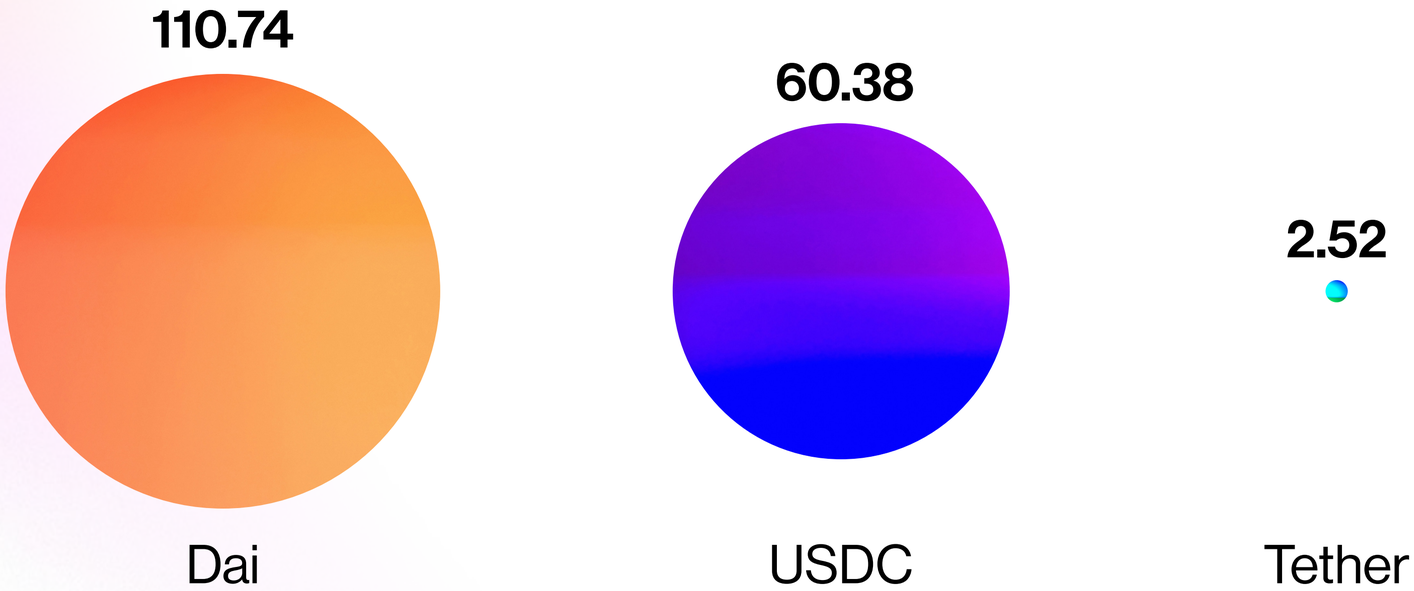
The on-chain velocity (i.e., rate of turnover) for today's top stablecoins *varies drastically*

Velocity
(annual on-chain transaction volume per unit of supply)



Source; Coin Metrics; Data is as of 5/12/2022.

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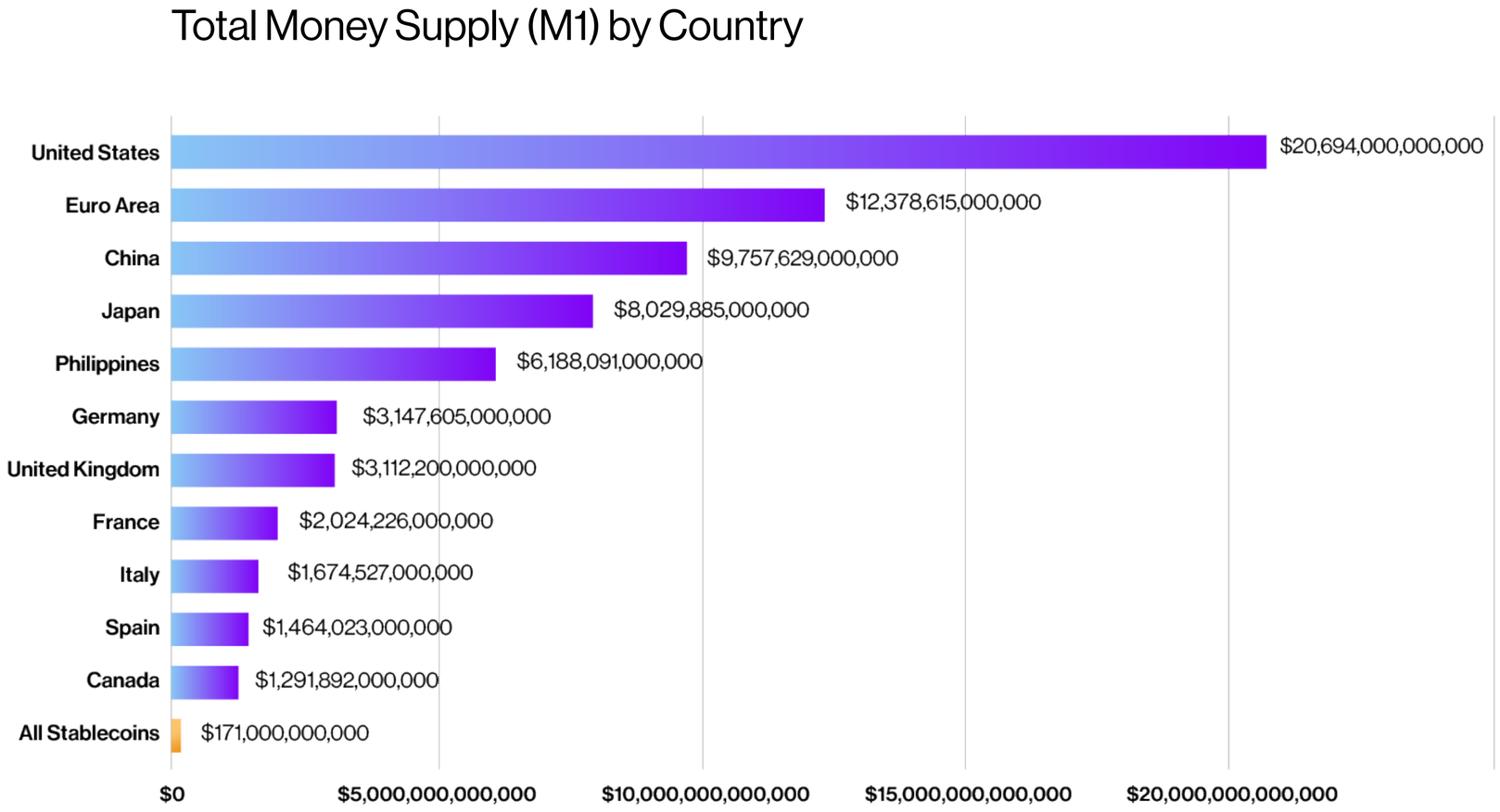
Stablecoins represent just a *tiny fraction* of the world's money supply



Source: Messari, https://take-profit.org/en/statistics/money-supply-m1/
Data is as of 5/12/2022.

NFTs

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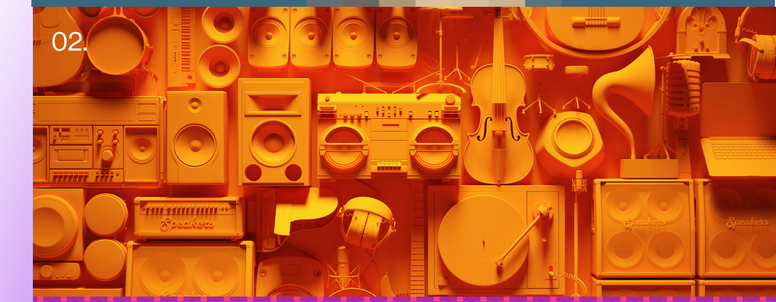
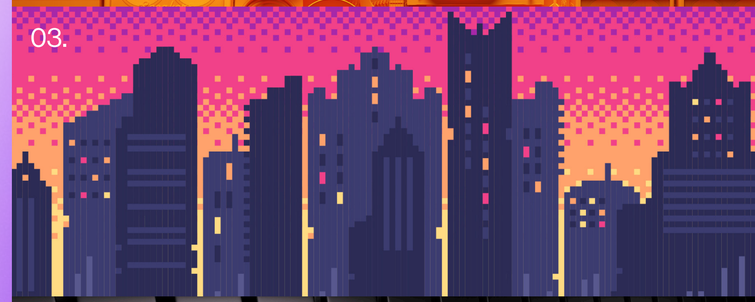
We've seen several categories of NFTs emerge, *but it's still very early days*

01.	02.	03.	04.	05.
Profile Pictures Images displayed on social media that represent an online identity	**Art & Music** A more efficient and transparent way for artists to monetize their work	**Gaming** Unique assets designed to be used within a game	**Utility** Items that grant special access, benefits, rewards, or memberships	**Virtual Worlds** Ownable properties that exist in new virtual worlds





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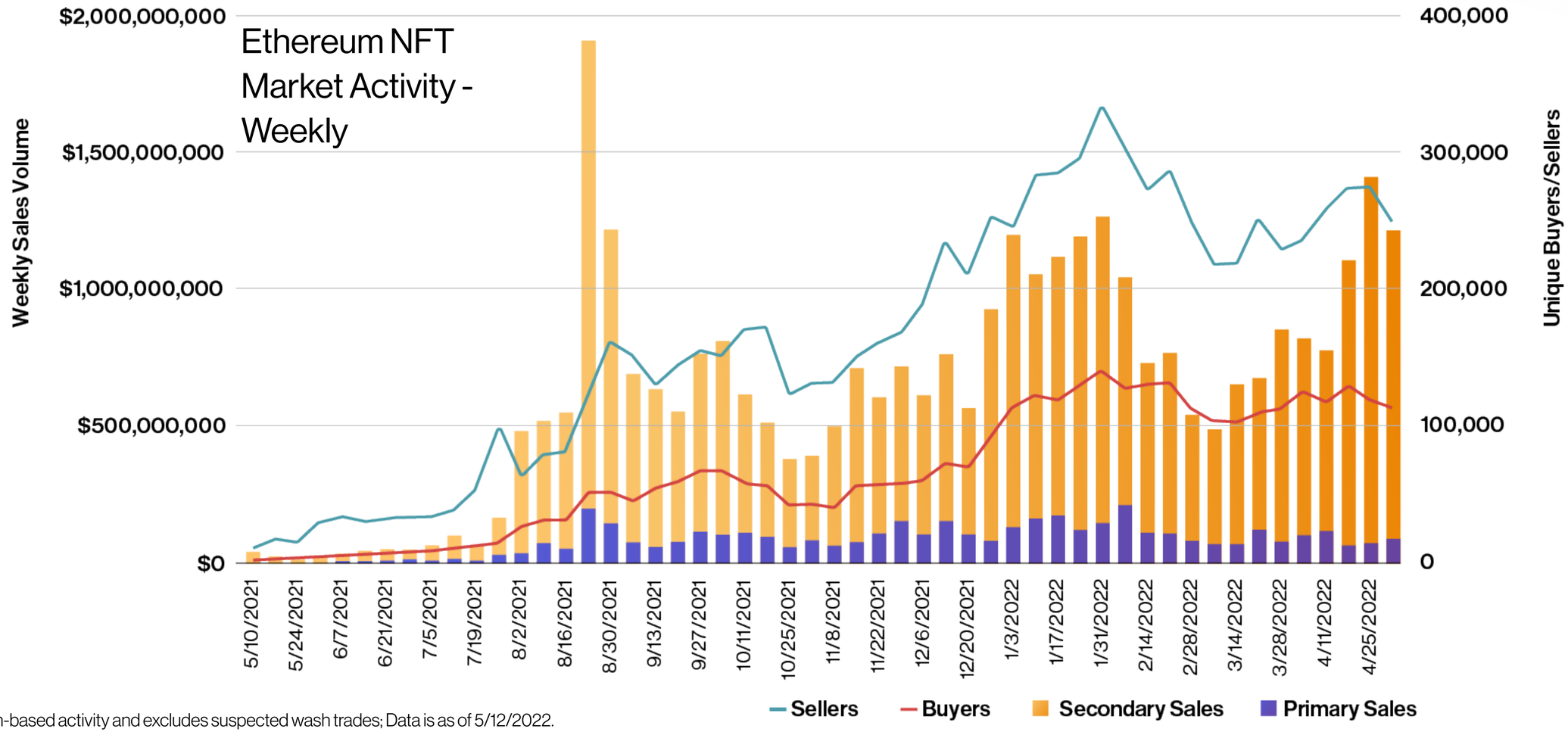
The NFT market *remains hot*, and we are beginning to see mainstream adoption



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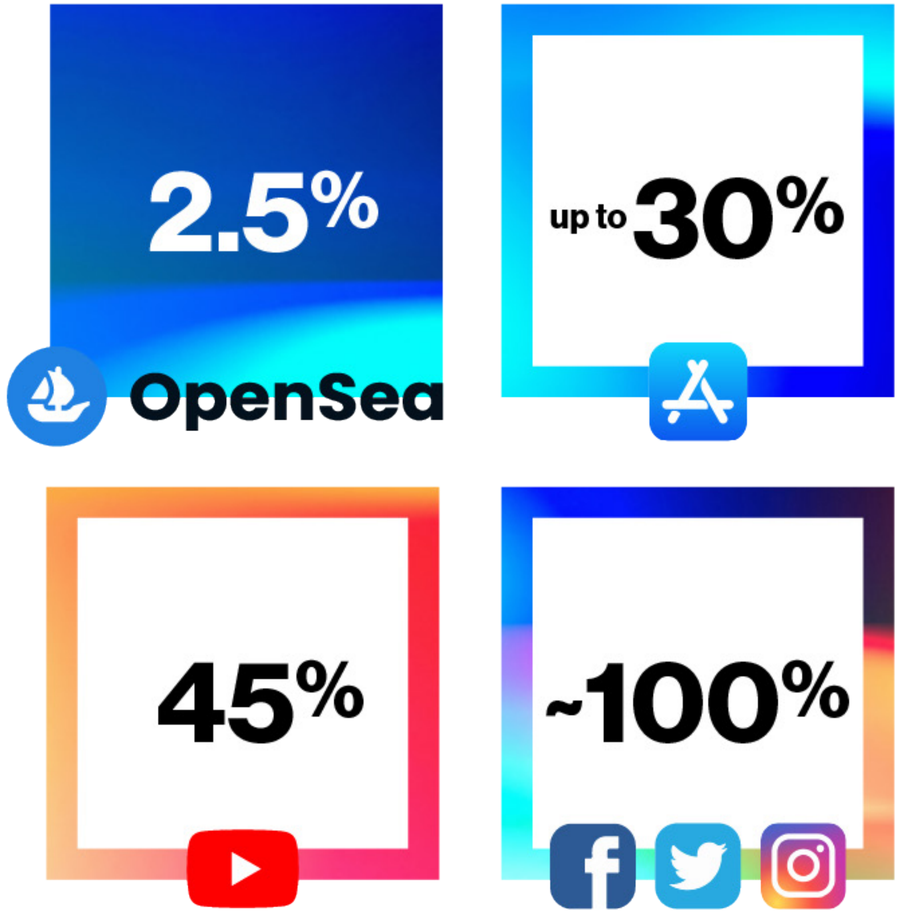
Source: Nansen; Includes all Ethereum-based activity and excludes suspected wash trades; Data is as of 5/12/2022.

Web3 platforms have *drastically lower take rates* **than the internet giants today**

"You know something is profoundly wrong with our economy when Big Tech has a higher take rate than the mafia."

– **Ritchie Torres,** U.S. Congressman representing the South Bronx

Take Rates of Popular Creator Platforms



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Estimated Creator Revenues by Source (2021):

*Includes ETH denominated primary sales of ERC-721 and ERC-1155 NFTs, plus creator royalties from secondary sales on OpenSea

*Amount paid to rights holders (e.g, record labels, distributors, etc.)

*Estimated based on YouTube's reported 55% of ad revenue distributed to creators

*Estimated based on Mark Zuckerberg's pledge to pay creators $1 billion through 2022

NFTs offer creators *a whole new way* to monetize directly with their fans

Sources:
1 a16z analysis of public data on the Ethereum blockchain,
2 https://www.forbes.com/sites/marisadellatto/2022/03/24/spotify-says-it-paid-7-billion-in-royalties-in-2021-amid-claims-of-low-pay-from-artists/
3 https://www.socialmediatoday.com/news/youtube-generated-288-billion-in-ad-revenue-in-2021-fueling-the-creator/
4 https://www.facebook.com/zuck/posts/10113607549897721

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There is *incredible innovation* happening in the NFT space

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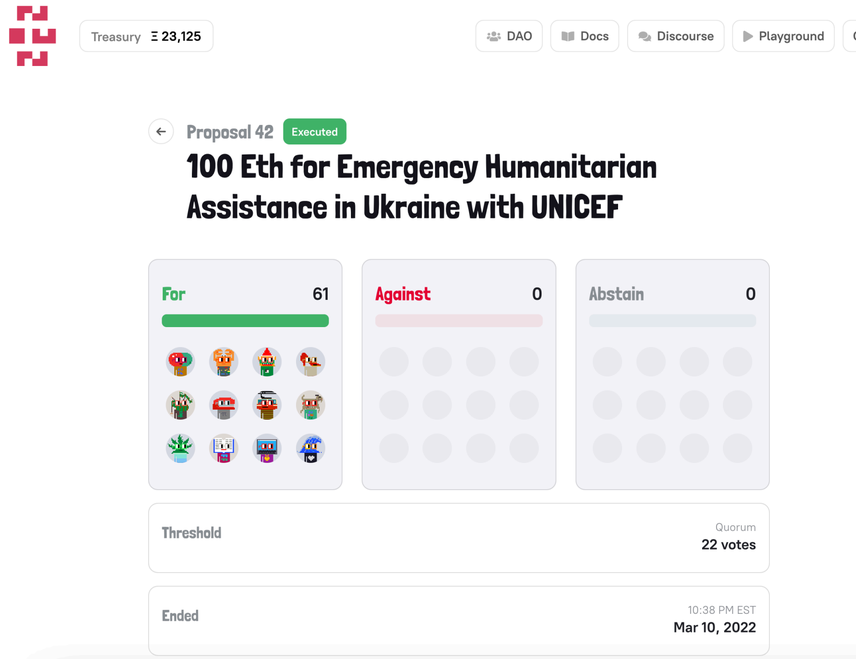
Airdrops

Music Royalties

IP/Commercial Rights

NFT DAO Governance



IRL Activation

Web3 Gaming

2022

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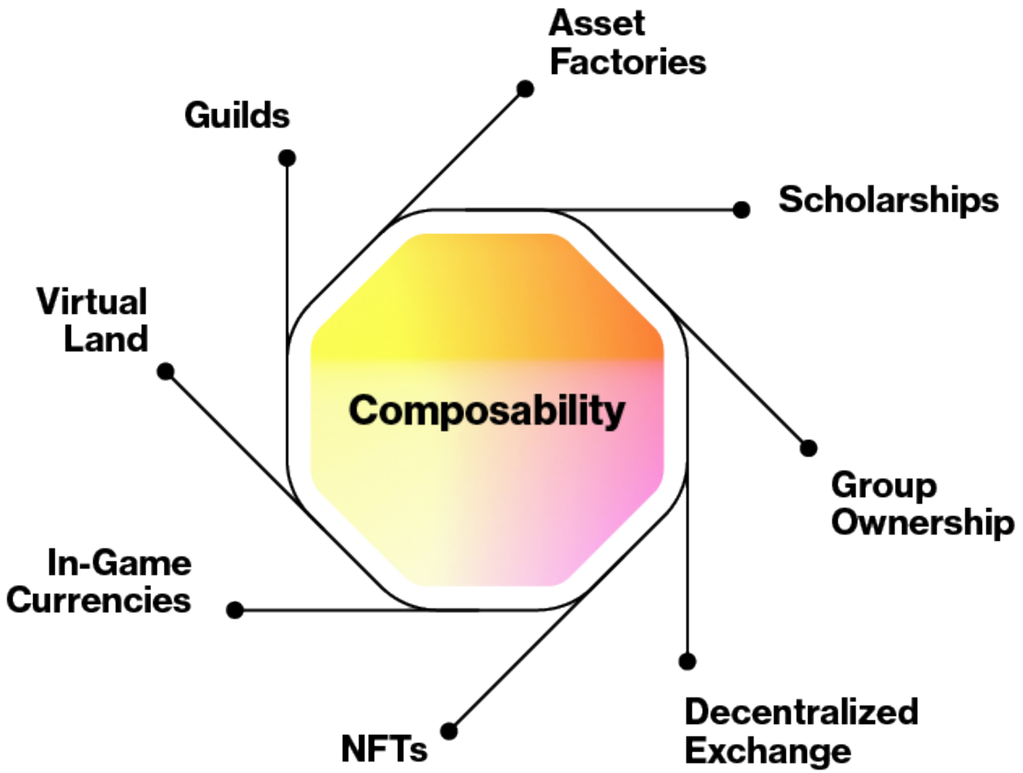
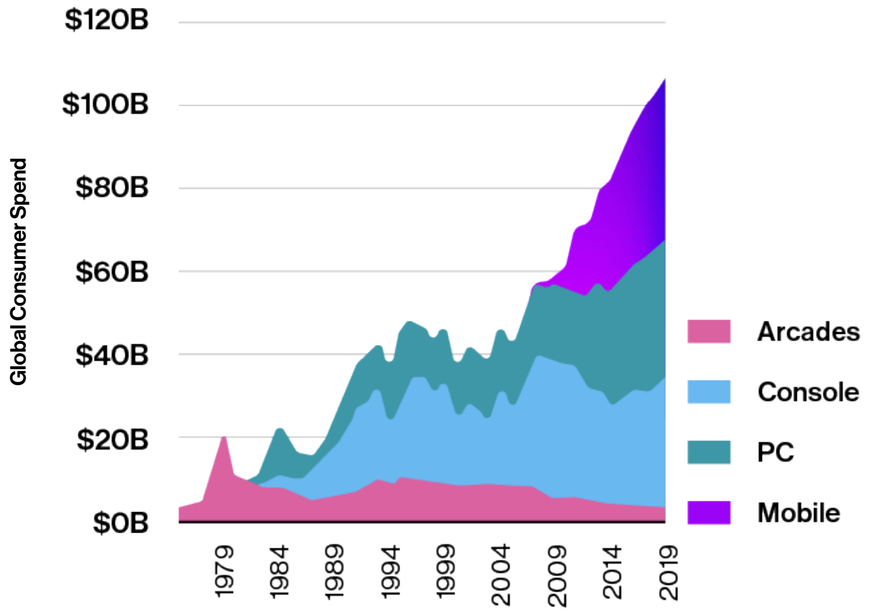
We are just beginning to see the *potential of web3 gaming*



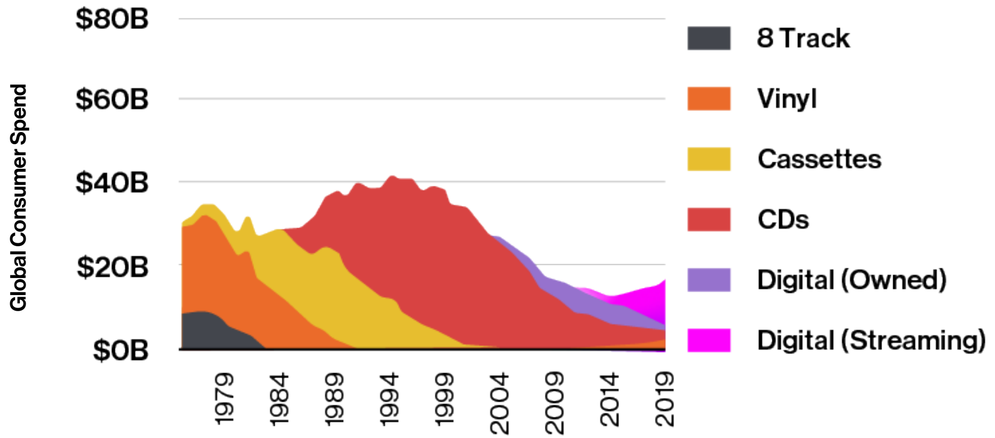
The video game industry has seen massive growth, driven by the *layering of new platforms and business models*

Video Games



When the internet came along, video games responded by embracing new genres like MMOs and battle royale, new business models like free-to-play and virtual goods, and new complementary behaviors like streaming.

Recorded Music



The music industry responded by spending decades filing lawsuits against innovators, and eventually accepting the incremental improvement of allowing users to subscribe to bundled content via streaming platforms.

Source: MatthewBall.vc

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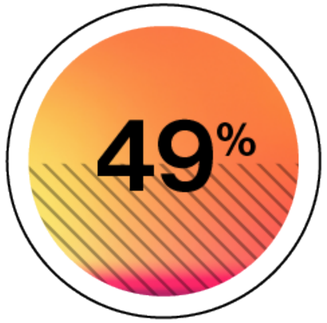
There is a *huge market opportunity* at the intersection of web3 and gaming

of NFT sales volume in 2021 was gaming-related assets



of crypto wallet activity comes from games

live web3 games with more than 1K unique on-chain users

Source: Blockchain Game Alliance, DappRadar; First two metrics are as of 12/22/2021, third metric is as of 5/12/2022.

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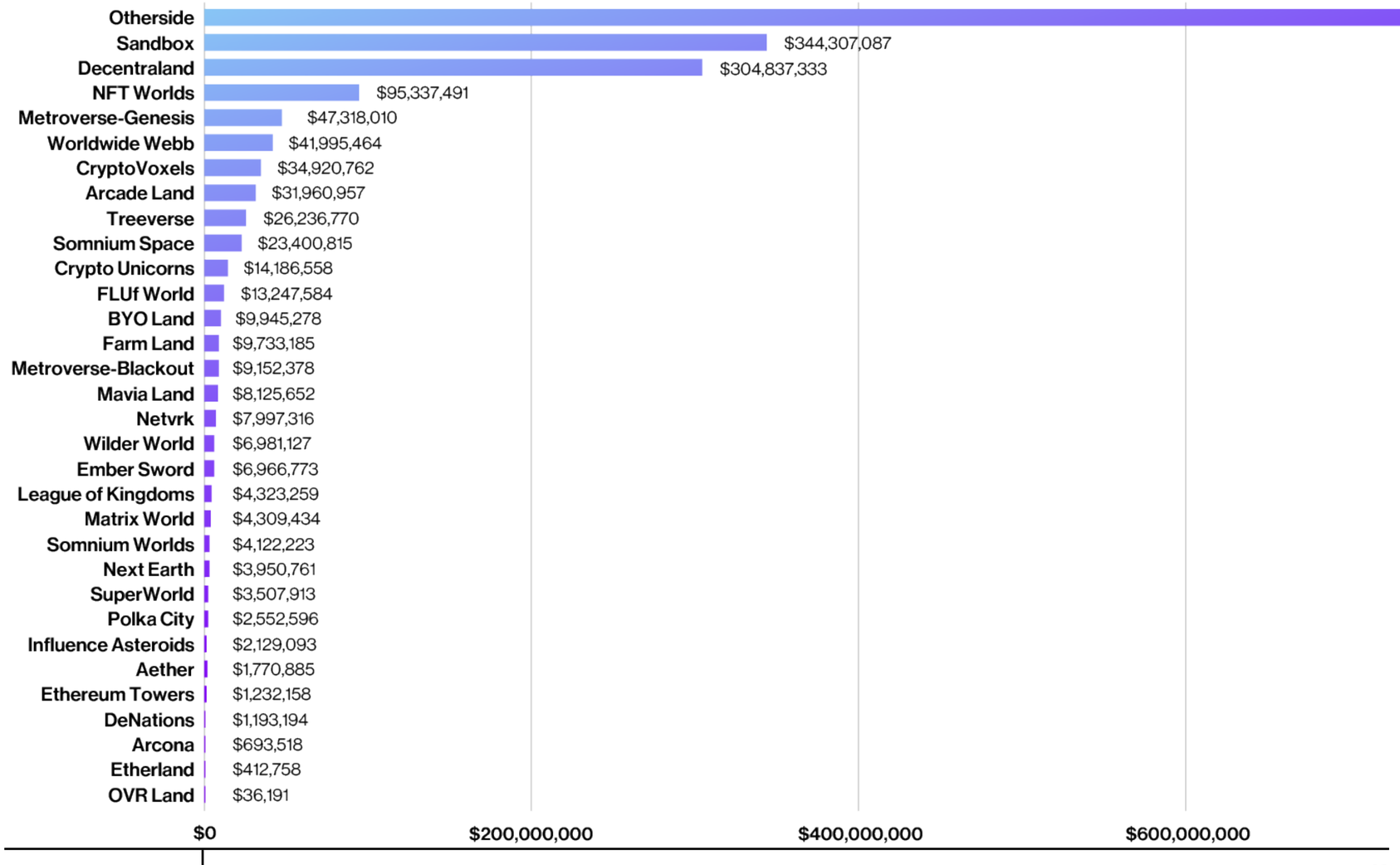
Metaverse Land—Secondary Sales



Web3 Gaming

Metaverse Land Sales

$1,808,580,704

Source: https://dune.xyz/metaland/Metaverse-Land-Community
Data is as of 5/12/2021.

Interest in the metaverse has exploded, **with land sales totaling nearly $2 billion**

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DAOs

2022

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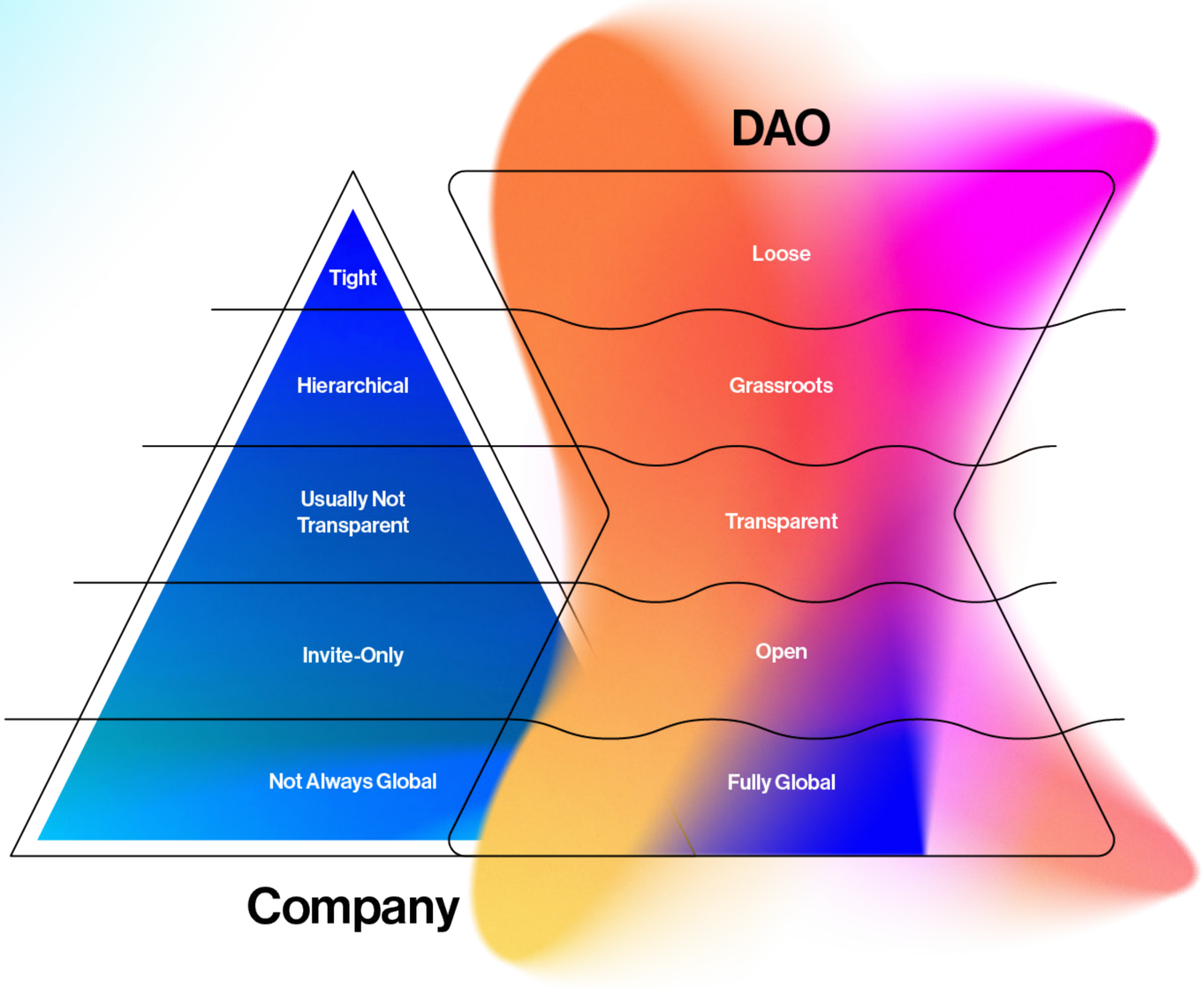
DAOs are providing a path toward *community governance* of any kind of organization



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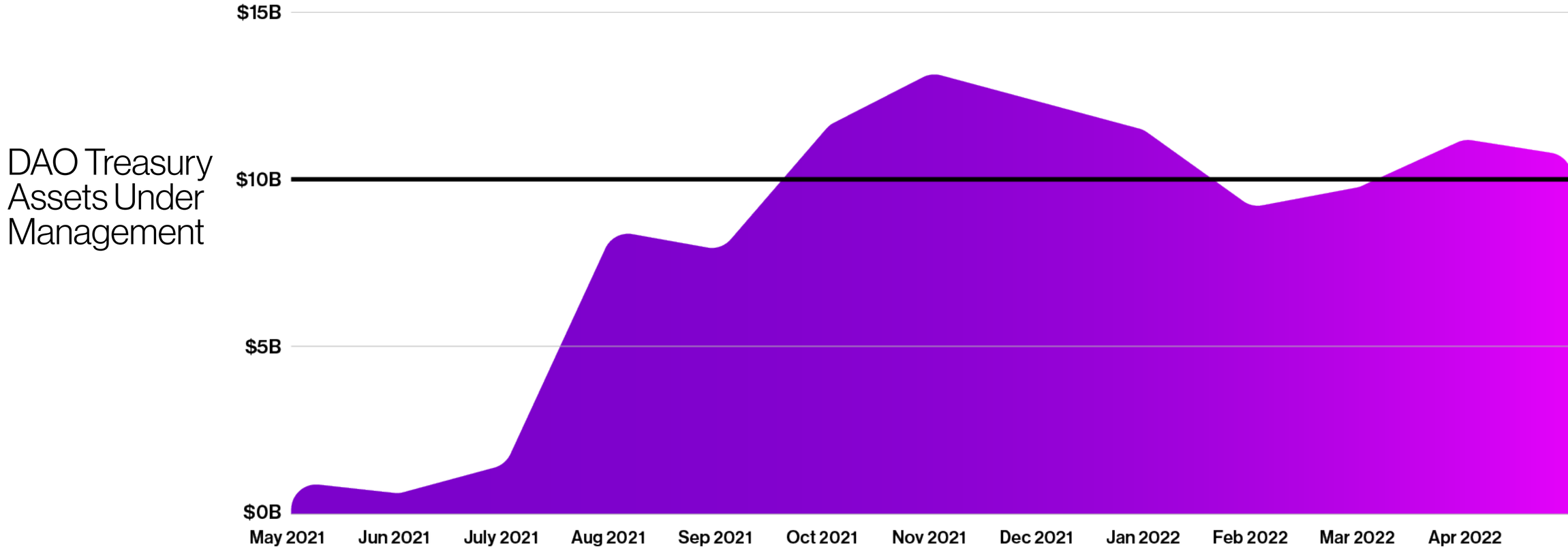
Source: https://deepdao.io/; Data is as of 5/12/2022.

DAOs have accumulated *more than $10 billion* in treasury assets under management



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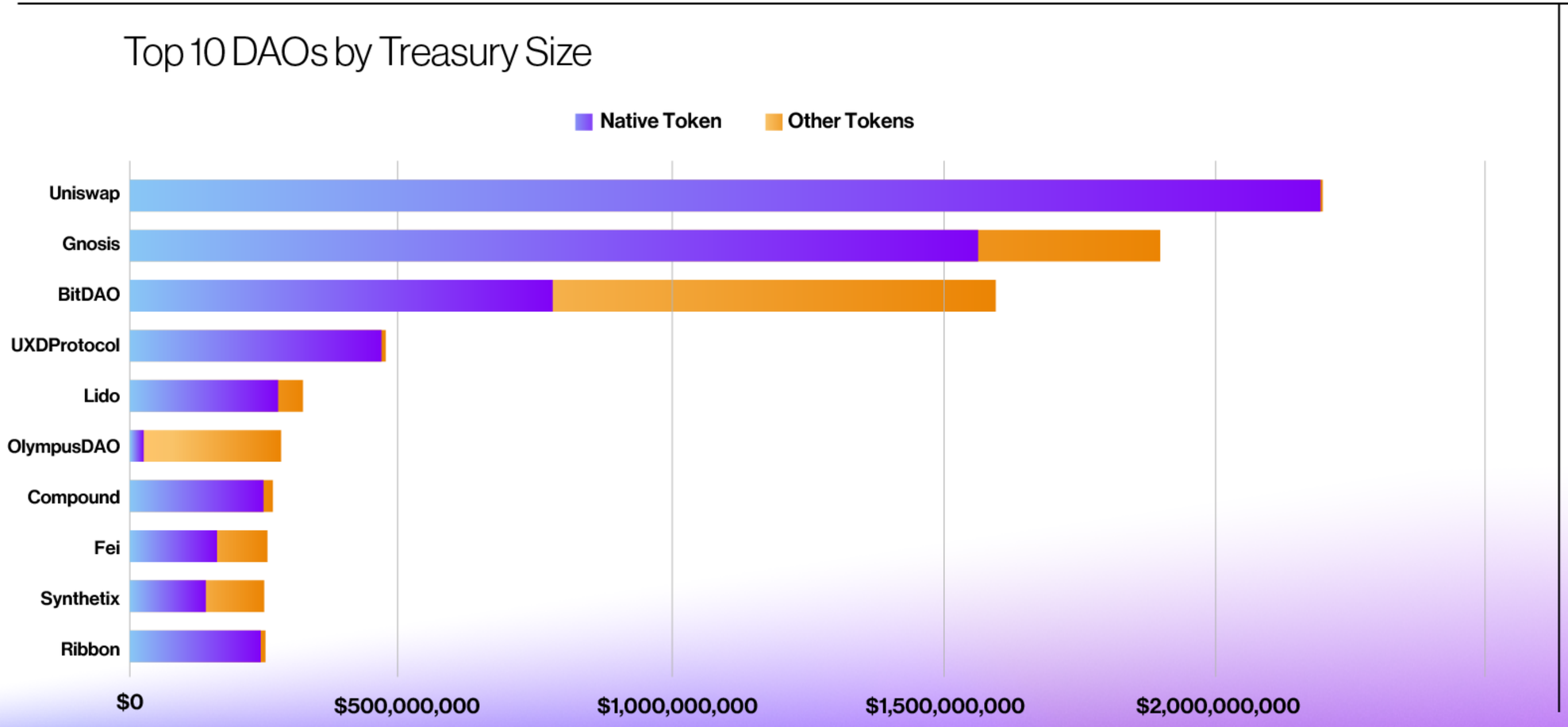
A large majority of value held by DAOs is in the form of the *protocol's native governance token*



DAOs

50

Source: https://deepdao.io/;
Data is as of 5/12/2022.

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DAOs enable *a new, bottom-up way* **for groups to make decisions**

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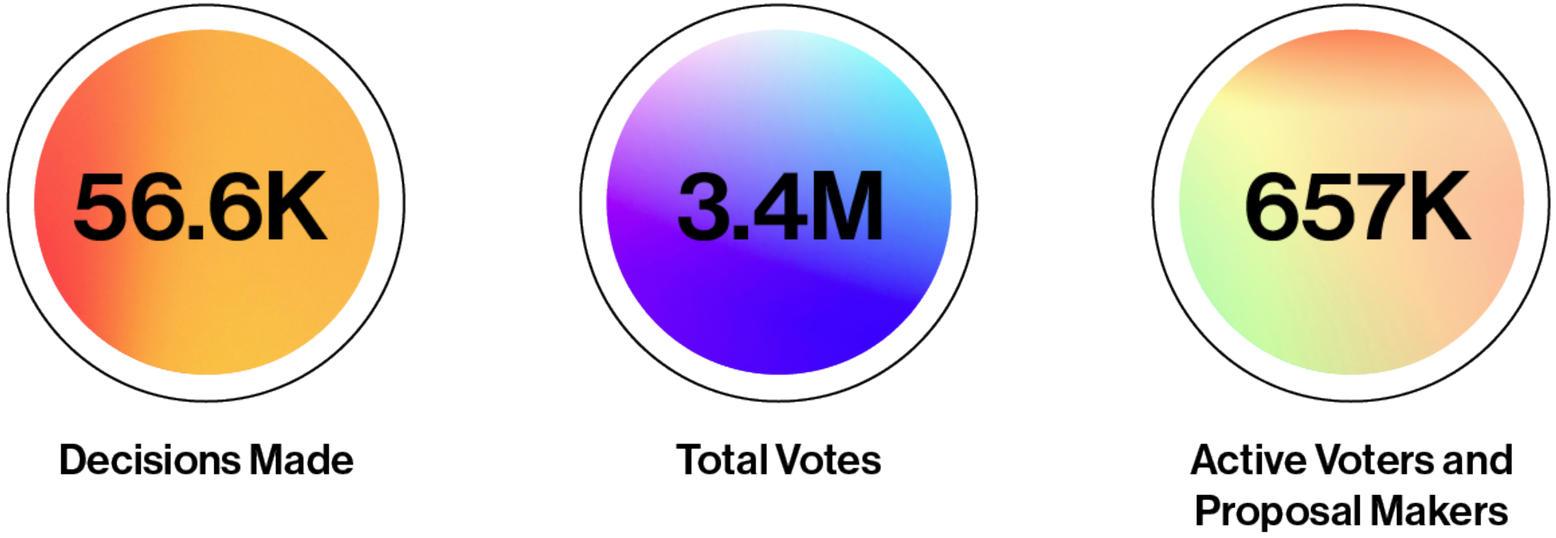

Source: https://deepdao.io/, Data is as of 5/12/2022.

Closing *Thoughts*

2022

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Crypto is already making a *real world impact*

Goldfinch

01.

Today, there is vast untapped lending potential in emerging markets. Goldfinch expands access to capital using crypto to truly empower financial inclusion.

Spruce

02.

Traditional web services rely on trusted intermediaries (Facebook, Google) to manage their users' digital identities. With Spruce, users can control their own identity with an Ethereum account.

helium

03.

The telecom industry is currently controlled by monopolies. Helium is building towards a future where network coverage becomes a commodity, fueled by competition, available anywhere in the world, at a fraction of current costs.

SOUND

04.

Only 15,000 musicians on streaming services (less than 0.2%) make more than $50K/year. Sound cuts out the middlemen, allowing artists to monetize directly from their fans.

flowcarbon

05.

The voluntary carbon market lacks liquidity, accessibility, and price transparency. Flowcarbon is leveraging blockchain tech to scale climate change solutions.

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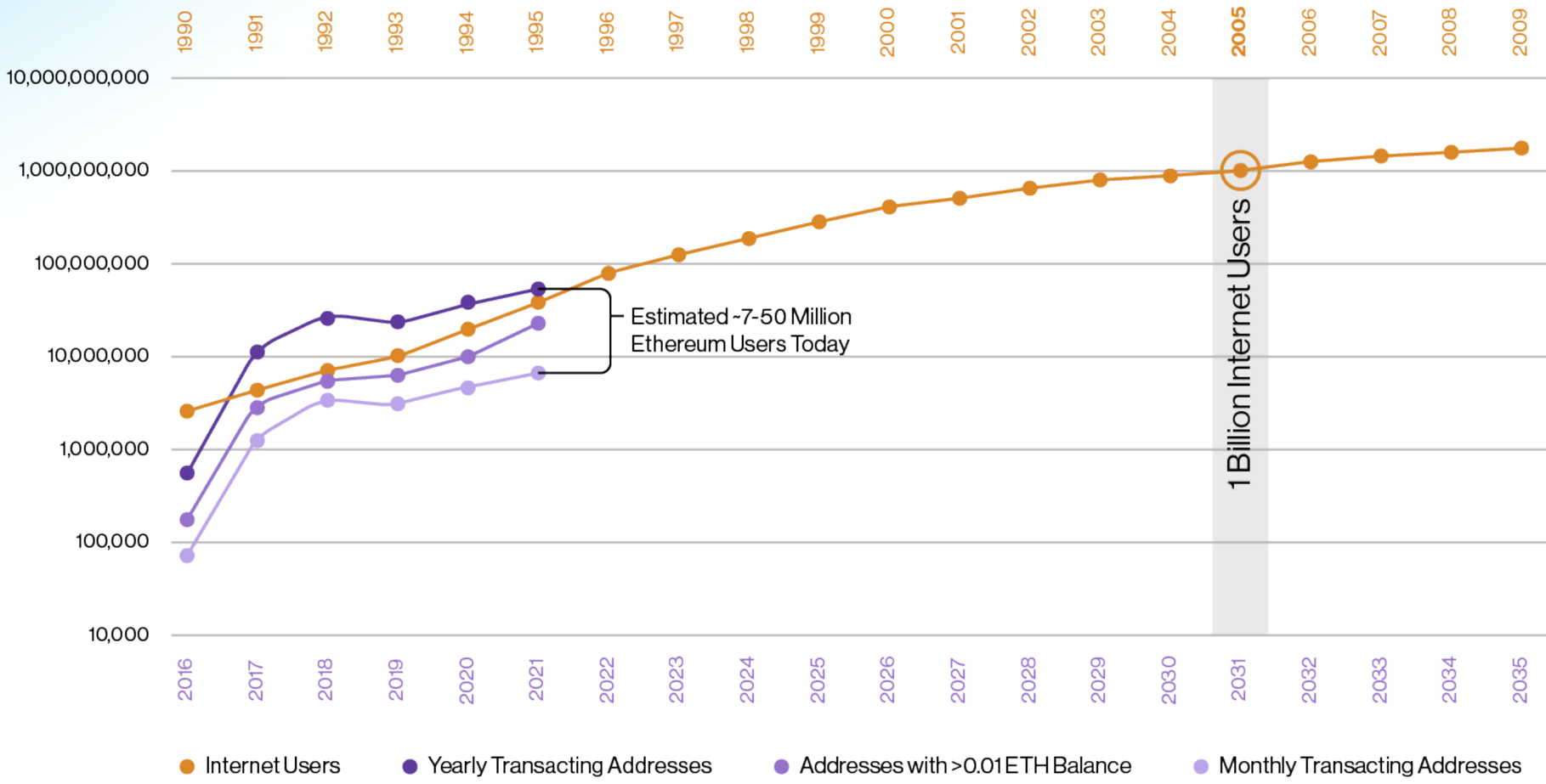
Source: World Bank, Coin Metrics, and a16z analysis of public data on the Ethereum blockchain
Data is through 12/31/2021.

It is still *early days* for web3

Internet Users vs. Ethereum Addresses (Log Scale)



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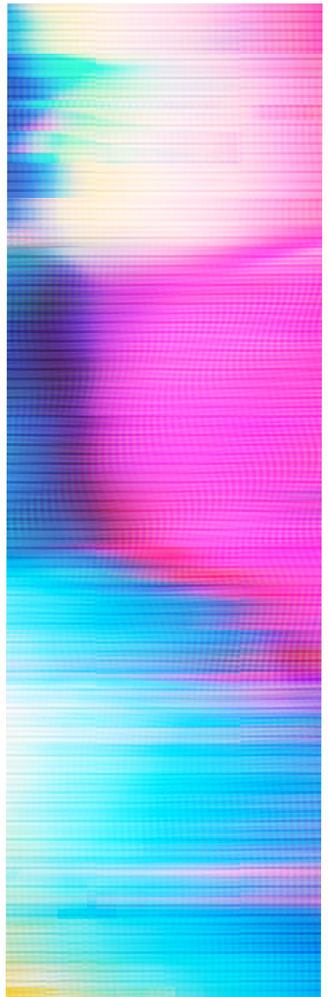
What's next **for web3?**



01.

Web3 infrastructure will continue to improve, with the continued improvement and growth of non-Ethereum L1s and Ethereum L2s. The Ethereum merge will hopefully happen this year. This will improve performance and also remove the environmental objection to using Ethereum.

02.

Expect many more creators to embrace NFTs in their own communities as they discover how much better they can monetize by selling digital goods directly to their fans vs banner ads and other web2 monetization methods.

03.

Expect many web3 games to launch, developed by founders with backgrounds at top game studios. Some of these will launch in the coming months, others will take longer.

04.

New web3 users who earn tokens through games and NFTs will likely choose to save those tokens in DeFi protocols vs tradfi banks given the better rates and user experience, growing the number of DeFi users.

05.

Expect many more experiments with DAOs, social tokens, decentralized social networks, web3 media, and other emerging categories. Some of these might break out in the same way DeFi, NFTs and web3 games did.

Source: https://cdixon.mirror.xyz/CGrfc6YezvzWXuOEShNvh46y4H4XTQvgYXNkFaZG510

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